# Cogito, Ergo Feci:* False Success and the Limits of Weight Surgery in a Computer-Use Agent

**Anonymous Author(s)**
Affiliation
Address
email

## Abstract

Computer-use agents often fail by announcing that a task is finished when it is not. We diagnose it in Holo-3.1-4B on OSWorld and test whether editing activations or weights, without training, removes it. The agent scores 56.7% on 119 diagnostic tasks; a false "done" is 57% of its failures, and re-sampled it answers "done" on most false completions too (72%, against 94% on true ones). With an input-exact replay and a gate fixed in advance, we find three traps that would each have produced a positive result: a false-success direction that adds at most 0.04 AUROC to position, run-to-run noise under which only 50% of false successes recur, and a loop direction in a smaller model that beats position yet reduces loops only at a large cost to healthy actions. No steering arm or weight edit passes the gate in pilot screens, which detect only a fall from 8/8 to 4/8 false answers and hold a single feasible false-success context, and no edited or prompted arm run online beats the run-to-run noise of the original. The tool call leans on a thought that already declares success: on 39 feasible contexts, neutral text in its place removes 80–82% of the `answer` margin, yet `answer` stays preferred in 37, perhaps carried by the success claim in the retained reasoning. Of the edits that move the call, a model merge shows no detectable selectivity (+0.7 points, 95% CI −16.1 to +16.3) and a closed-form edit lowers false answers more, with an interval that includes zero. An external verifier that checks each requirement on the final screen separates false from true successes offline, detecting 68% of false successes and rejecting 10% of true ones (the agent judging itself with a free-form prompt: 53% and 48%); online, on 120 held-out tasks under a rule fixed in advance, it lowers the false "done" rate from 22.5% to 10.8% (−11.7 points, 95% CI −19.2 to −4.2) with no measurable change in score (+1.6, −4.4 to +7.7).

## 1 Introduction

Computer-use agents run a closed loop: read a screenshot, act, repeat until they declare the task done or run out of steps. A recurring way to fail is to declare it done when it is not. The agent announces success, the episode ends, and the environment's checker returns zero [25]. Figure 1a shows one such episode: the agent writes that it cannot finish without a password, then reports that the installation "has been initiated successfully". Users report the same failure in deployment, including an agent that keeps acting while its screen capture fails, then claims completion [6].

Interpretability offers a tempting repair. If a behaviour is mediated by one direction in the residual stream, projecting it out of every matrix that writes into the stream removes the behaviour without

*"I think, therefore I have done", after Descartes: the agent's reasoning concludes the task is done before checking.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

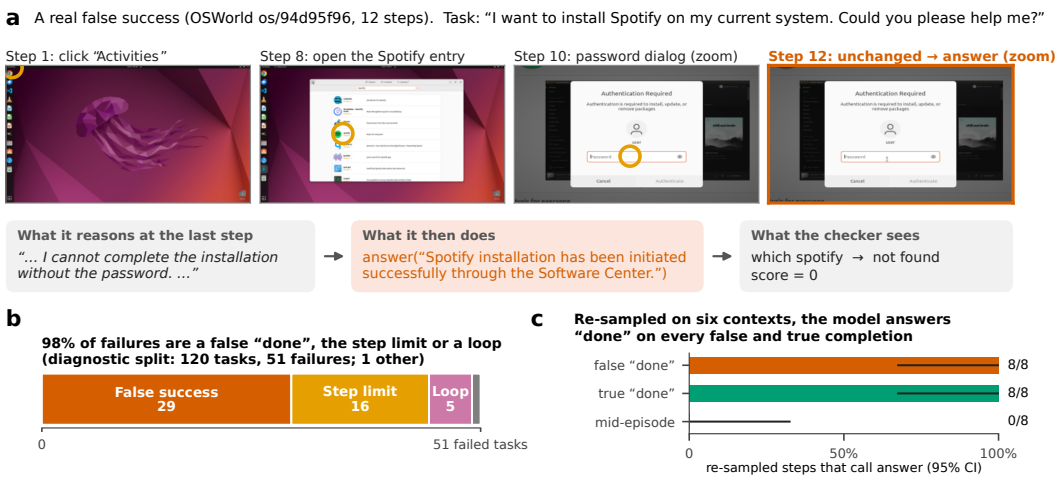



**Figure 1: The agent declares success it has not reached.** (a) A real episode: the reasoning says “I cannot complete the installation without the password.”, the next action claims success, the checker scores 0. (b) Failures on the frozen diagnostic split. (c) Re-sampled from the same six contexts ($k = 4$), the original model calls `answer` on 8/8 false-completion and 8/8 true-completion samples, and on 0/8 mid-episode samples; bars show Wilson 95% intervals; one false-completion context is an infeasible task (Appendix M).

a gradient step [2], and “weight-space programming” recently removed a looping habit from a 0.8B model by editing 131,072 weights [31]. False success looks like an ideal target: it is frequent (Fig. 1b) and it is a single discrete decision, the call to the `answer` tool, of the kind single directions are known to steer [4, 36]. This paper is about the gap between finding such a direction and proving that it causes the behaviour, and about what happens when one tries to edit the behaviour away anyway.

**Contributions.** We report only what we measured.

1. **A diagnosis of a real agent on real desktops** (§2). Holo-3.1-4B solves 66/119 tasks of a frozen OSWorld split; 29 of its 51 failures are false successes, and re-sampled, it still says “done” on most false completions (72%, against 94% on true ones).
2. **Three measured traps and the protocol they impose** (§3): position-confounded directions, regression to the mean between runs, and detection without causation. Each would have produced a false positive here.
3. **A causal record of every intervention we ran** (§4), from steering to mechanism-level weight edits and model merging, as pre-registered pilot screens with a semantic audit of every changed action. None passes the gate, and the screens could find only large effects: the gate flags a fall to 4/8 false answers or fewer, and one of the two false-success screening contexts is an infeasible task, so the evidence on feasible false successes rests on a single context and a few samples per arm.
4. **A mechanistic account of the failure to edit** (§4.1, §4.3): the tool call leans on a thought that already states completion, and of the two edits that move it, a merge shows no detectable selectivity on a strictly feasible cohort, while the selectivity of a closed-form edit is not established, in line with probes that separate the two no better than position. No arm run online beats the run-to-run difference of the original (§4.4).
5. **An external verifier** (§4.4, Appendix G): offline it detects 68% of false successes at 10% false rejections, where the agent judging itself with a free-form prompt does not separate them; online, in one TEST run under a rule fixed in advance, it halves the false “done” rate with no measurable score change.

## 2 Setting and diagnosis

**Models and harness.** Holo-3.1 models are fine-tunes of the Qwen family [9], released as Qwen3.5 models with a hybrid decoder (three gated DeltaNet layers per full-attention layer, tied embeddings); the 4B has 32 layers of width 2560. We reuse the Holo agent loop as packaged by RealtimeGUI

[1], a third-party reproduction whose infeasibility rule is broader than the official adapter's (absolute scores are not strictly comparable with H Company's; paired comparisons are), served by vLLM 0.23 [17] with thinking enabled, $T = 0.8$, at most 50 steps and the three latest screenshots. OSWorld [38] tasks run unmodified in Docker+KVM VMs. From 361 tasks we freeze a 120-task DIAG split and a 120-task TEST split that no construction or screening read; 5 online runs and the guarded agent used it (§4.4; Appendix B).

**Most failures are a false "done".** Each episode gets one category from rules over its actions, screenshots and score, hashed before any direction is fitted: a *false success* calls `answer` on a feasible task scored 0 or claims completion on an infeasible one; FS+ marks such a final `answer` step and FS− that of a true completion. Holo-3.1-4B scores 56.7% (official OSWorld score, partial credit included, the convention for every score here) and fully solves 66 of 119 tasks (55.5%, 95% CI 46.5–64.1). Of 51 failures, 29 (57%) are false successes, 16 have the step limit as primary category and 5 loop. False successes occur in every domain but one and are not early quits (median 18 steps; Appendix C).

**The model says "done" on most false completions too.** Re-sampled on six screening contexts ($k = 4$, production sampling), the original model calls `answer` on 8/8 false-completion and 8/8 true-completion samples, and on 0/8 mid-episode ones (Fig. 1c); on the strictly feasible cohort of §4, it answers on 55/76 (72%) false and 75/80 (94%) true completions. 9 of the false successes are infeasible tasks on which the model claimed completion; there, `answer` with a refusal is the correct tool, so these contexts are flagged wherever $P(\texttt{answer})$ is counted (Appendix M); yet 8 of 10 final answers on infeasible tasks claim success, so they are mostly false successes, not misworded refusals. Every re-sampled reasoning trace at a completion step opens with a success claim at generated token 4–5, with no verification word before it, on true completions as on false ones (0/16). Any repair must change a decision taken in most cases of both kinds, in only one of them.

## 3 Three traps, and the protocol they impose

We never record activations during rollouts. Each step's exact chat is rebuilt and replayed in `transformers`; the rebuilt messages match those sent on 7/7 steps, the prefill length matches vLLM's on 12/12, and the replayed model gives its own actions a mean log-probability of $-0.16$ per token against $-1.10$ for synthetic ones (Appendix D).

**Trap 1: the direction measures position.** False completions come later in an episode than true ones. A first FS+/FS− difference-of-means direction reached AUROC 0.836, against 0.870 for the step index alone. The pre-registered analysis (41 FS+ and 109 FS− steps, 95 tasks, task-grouped cross-validation) finds a best direction at 0.76 against 0.724 and 0.735 for the step and context-length baselines; a logistic model on position alone scores 0.722, and adding the best direction raises it by at most 0.04 (Fig. 2b, Table 1). We find no false-success signal beyond the length of the episode.

**Trap 2: regression to the mean.** Re-running the unmodified agent on the same tasks, only 9/18 (50%, 95% CI 29–71) pass-1 false successes recur (Fig. 2c). A patch evaluated on "false-success tasks before and after" would be credited with about 50 points of reduction while doing nothing.

**Trap 3: detecting is not causing.** On Holo-3.1-0.8B in MiniWoB++, a loop direction reaches task-grouped AUROC 0.893, far above 0.70 for the step index (Fig. 2a), and is a poor lever: erasing it from every writer leaves loop actions as likely as before, its best weight edit is less selective than a random direction, and steering along it reduces loops only at a large cost to healthy actions (§5).

**Protocol.** (i) Free generation is the only efficacy measure: each candidate regenerates the full reasoning ($k = 4$, seeds shared across arms), and invalid outputs stay in every denominator. (ii) A gate fixed before any data: on FS+ contexts of tasks not used to build the intervention, $P(\texttt{answer})$ must fall by at least 25 points and 50% relative, while the FS− answer rate and valid healthy actions may fall by at most 5 points; the FS− criterion separates "less false success" from "less success". (iii) Random directions of matched norm and scope, identity hooks and a budget of generation batches fixed before execution. (iv) A confirmation reserve excluded from every construction and screening, kept to confirm a committed candidate; since none was admitted, five non-admitted arms

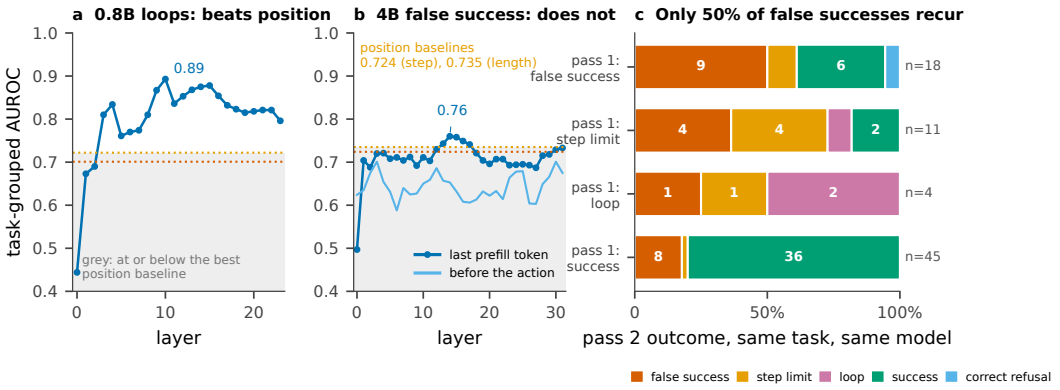



**Figure 2: Two of the three traps.** (a) Holo-3.1-0.8B, MiniWoB++: a loop direction separates loop steps far above the step-index (red) and context-length (orange) baselines, with task-grouped cross-validation. It still fails every causal test (§H). (b) Holo-3.1-4B, OSWorld, official analysis on both passes (41 FS+, 109 FS−, 95 tasks): the false-versus-true completion direction barely clears the position baselines at its best layer, and not once position is controlled (Table 1). The grey band marks AUROCs at or below the best baseline. (c) The unmodified 4B run twice on the same tasks (80 tasks scored in both passes). Only 9 of 18 pass-1 false successes recur, against 36 of 45 pass-1 successes.

| Predictor (task-grouped 5-fold CV) | Best layer | AUROC |
|---|---|---|
| Step index alone | – | 0.724 |
| Context length alone | – | 0.735 |
| Direction, last prefill token | 14 | 0.760 |
| Direction, before the action | 3 | 0.701 |
| Position (step, log length), logistic | – | 0.722 |
| + direction, last prefill token | 15† | 0.762 |
| + direction, before the action | 22† | 0.717 |

**Table 1: Official false-success direction analysis: no signal beyond position.** Holo-3.1-4B, 41 FS+ and 109 FS− steps from 95 tasks, two passes, 5-fold cross-validation grouped by task, labels frozen before analysis. Exploratory: the second pass stopped at 80 tasks. †Layer chosen after the fact.

were run on it instead (§4.4), a departure we list as a limitation. Every changed action is read by a blind semantic audit, because a moved tool call is not a correction (§4.3).

# 4 No selective effect found

## 4.1 Where the decision is carried

Patching the residual stream from the same task's previous step (a click) at each layer and four positions, the decision is invisible at the end of the prefill and appears only at the tool-name position from about layer 16 upward (Fig. 3). Three attention heads, which read the system-prompt definition of `answer` and the JSON being written, and ten late MLP neurons move the margin, the same way on true and false completions; removing all of them lowers it by 11.2–12.2 logits and flips 0 of 4 decisions. At the tool-name position the margin of `answer` over the best other tool is 13.9–19.1 logits on false completions and 10.9–17.8 on true ones. That position follows the JSON `thought` field, which already states that the task is complete, so the margin measures confidence in carrying forward a verbalised decision rather than the margin of the decision.

A ceiling test asks how much must be replaced at once (Fig. 4). With the recorded input forced, whole blocks of writer outputs are swapped in at the decision position from a donor step where the model clicked. On the first FS+ context (margin +16.9), all 32 MLPs together bring the margin to +0.9 without flipping it, while the MLPs and token mixers of layers 12–23 flip it (−3.5), as do those of layers 24–31 (−10.0); a second FS+ context (−3.0) and a true completion (−2.2) flip the same way. Self-swap and residual swap reproduce the baseline and the donor exactly. The first context

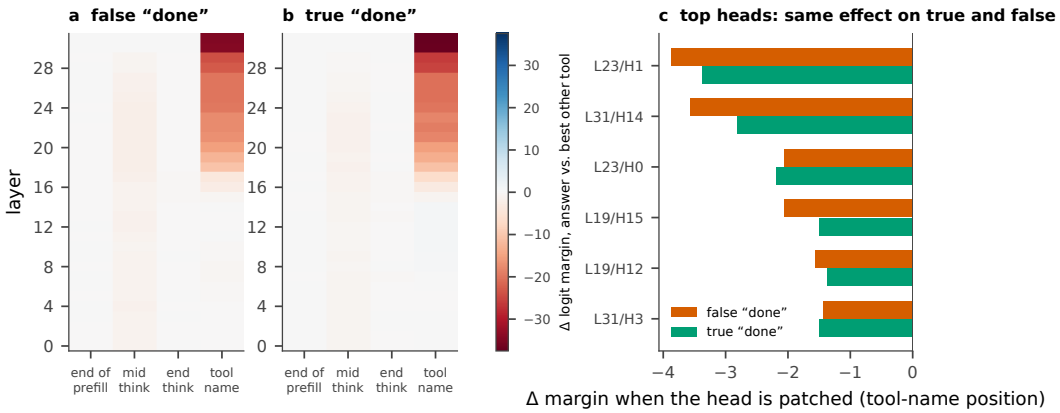



**Figure 3: The tool choice becomes visible late, at the tool name, by the same components for true and false completions.** Activation patching on Holo-3.1-4B from the previous step of the same task (two tasks per class; exploratory construction data). (a, b) Mean change of the `answer`-versus-best-other-tool logit margin when the residual stream is patched at one layer and position; red lowers `answer`. (c) The six heads with the largest effect at the tool-name position; red bars are false completions, green bars true ones.

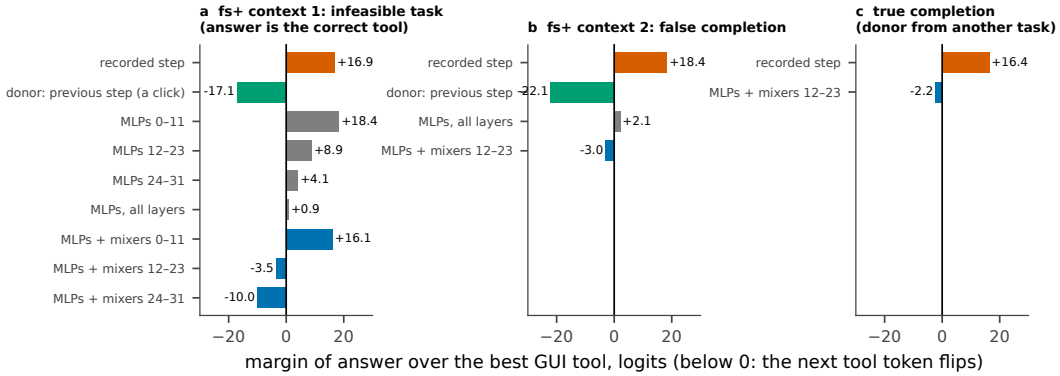



**Figure 4: Ceiling test: the next tool token flips only when MLPs and token mixers of a whole block are swapped, on false and true completions alike.** Holo-3.1-4B, recorded input forced, logits only. Bars give the margin of `answer` over the best GUI tool after swapping in, at the decision position, the outputs of the indicated writers from a donor step where the model clicked. (a) The first FS+ context is an infeasible task, where `answer` is the correct tool. (c) The true completion uses a donor from another task. Three contexts; exploratory.

is an infeasible task, where `answer` is correct; the donor differs in screen, thought and position, not only in decision; and since the true completion flips too, the test locates where the next tool token is assembled, not a mechanism of false success. Removing one direction of the swap difference from the writers of layers 12–23 brings both FS+ margins near zero (+1.4, +1.6) but also two true-completion margins (+4.8, +2.8). On the 19 false and 20 true completions of a strictly feasible cohort, teacher-forced with the screen, history and reasoning fixed, replacing the note and thought by a neutral text of the same token length removes 14.0 logits of margin on false completions (95% CI 12.9–15.1) and 14.5 on true ones (12.9–16.0), 82% and 80% of it (difference −0.46, −2.36 to +1.50), yet `answer` stays preferred in 37 of 39 contexts: the note reinforces the conclusion; the success claim left in the retained reasoning may carry the rest. A counterfactual “not complete yet” text nearly cancels the margin on average, and no probe beats the step index after correction (best 0.64 against 0.70; detecting AUROC below 0.725–0.755 was underpowered). A first two-context version with a donor’s text flips the decision (Appendix F.2). Blanking the inner tokens of the note and thought with spaces at fixed positions, a different neutralisation, lowers the margin by 7.55 and 8.18 logits, with no detectable difference between strata (−0.64, 95% CI −2.13 to +0.80); this smaller effect is partly carried by single DeltaNet layer outputs (layers 18, 22, 28 and 30, 0.9–1.6 logits each, not additive), while blocking any single head’s final-query attention costs at most 0.12 (exploratory, same cohort).

### 4.2 Steering and targeted edits

Table 2 lists every intervention on the six screening contexts, each testing a hypothesis from the literature: tool choice is linearly steerable [36]; the decision is taken during reasoning; false success is excess expected value or commitment [13, 15]; late MLP components write it [7], tested here at the neuron level; the model fails to check; false success is a perception of completion; the model opens with its conclusion; memory of the early episode fades; the circuit of §4.1 carries it. No arm passes the gate. Removing the whole circuit answers on every false completion, like its random control. Pushing toward low value makes an independent reader find an explicit intention to re-evaluate in 3/8 FS+ traces against 1/8 at baseline, with 0 verification actions and the same answer rate: the model writes that it should check, then answers. A 2560-weight edit of the output row of "Wait" lowers false answers to 6/8 with every control intact, short of the 50% criterion, and an audit rates none of its replacement actions useful. A conditional variant yields *intentions* to correct (the model states the right dates, then generates an unexecuted click on a summary that no evidence shows to be editable) and transfers to 0/12 useful actions on held-out tasks.

Each efficacy cell holds 8 generations from two tasks; 8/8 has a Wilson interval of [68, 100]%, so these screens exclude only large effects. The doses tested, 0.13 to 2 times the residual norm, may be too small to matter, and a ceiling effect is likely.

### 4.3 Weight edits: levers without established selectivity

Weight edits that need no direction or that target a mechanism (Appendix N) were served on separate hardware against a re-served original, with the same 8 seeds on 6 contexts. Extrapolating the fine-tune away from Qwen, $W(\lambda) = W_{\text{Qwen}} + \lambda(W_{\text{Holo}} - W_{\text{Qwen}})$, called ExPO [44], leaves false answers at 16/16 in all arms but one (15/16). The mechanism-level edits H2–H10 (memory gates, MLPs or all language tensors moved toward Qwen, the end-of-thinking embedding, the attention temperature, the rotary share of the query gain) leave the tool call in place, break generation, or move it only on the infeasible context, where theme clicks replace an answer that should have explained the impossibility; a blind audit finds no useful action (Table 7). Two-neuron conditional circuits, an adaptation of the weight-space programming recipe and an explicit instruction to check each requirement, injected into the reasoning, do no better. Two edits move the decision on feasible contexts. Merging a fraction $\beta$ of the task vector of CUA-Lite, a Qwen3.5-4B fine-tune on desktop data, lowers false-completion answers to 15/16 and 11/16 at $\beta = 0.25$ and $0.5$, but true-completion answers further, to 11/16 and 2/16 (on the feasible FS+ context alone, 3/8 at $\beta = 0.5$): the model continues instead of concluding, whatever the truth of the completion. Online, the merge shows no detectable gain over an original served by the same endpoint: +1.9 points over 165 tasks (95% CI $-4.3$ to +8.0; §4.4).

**A closed-form edit.** A closed-form edit of the 24 residual writers of layers 12–23, in the spirit of AlphaEdit [8] (a ridge update toward the donor displacement, projected off protected keys), moves the decision too: on contexts outside its construction, false-completion answers fall from 15/16 to 8/16 (all on one of two contexts) and true-completion answers from 15/16 to 3/16, with healthy actions intact (on a finished e-mail task, three replacements would write into an already correct field), while a random update of matched per-writer norm leaves 16/16 and 16/16: a real causal lever on stopping, unselective on these contexts. On a strictly feasible cohort of 19 false and 20 true completions (Appendix F.5), the merge lowers both with no detectable difference (selectivity +0.7, 95% CI $-16.1$ to +16.3), while the closed-form edit lowers false answers by $-17.1$ points ($-28.9$ to $-6.6$) and true ones by $-8.8$, a selectivity of $-8.4$ ($-22.6$ to +5.9) that is not established either way; its random control raises false answers instead (+15.8 points), as do zeroing four attention heads (+21.1) and adding an RL-trained donor's task vector (+7.9): a false "done" behaves like the default slope of a degraded model. An edit at or after layer $\ell$ can treat two states differently only to the extent that some function of the layer-$\ell$ residual stream separates them, and usefully only if that function generalises to unseen tasks. On the inputs of the 24 writers of layers 12–23, a task-grouped probe separates 19 false completions from 19 true completions no better than position: median AUROC 0.704 at the prefill and 0.634 at the end of reasoning against 0.852 for the same probe on the step index (within-fold, pair-weighted over 71 pairs), or 0.596 and 0.496 against 0.644 pooled out of fold; and 83–86% of the false-completion key energy lies in the true-completion subspace. But 5 of these 19 false completions are infeasible tasks, where `answer` is correct, so the probe mixes in cases with no wrong decision, which weakens the argument; the strictly feasible cohort above avoids this for

| Hypothesis & intervention | Where | Dose | Out. | fs+ `answer` [95% CI] ↓ | fs− `answer` = | healthy valid = | invalid | Verdict |
|---|---|---|---|---|---|---|---|---|
| Original (reference) | – | – | – | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | – |
| `4b-x`: residual addition, matched fs direction | L16, last prefill, $\alpha = -1.20$ | 0.13 | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | rejected |
| `4b-x`: residual addition, matched fs direction | L24, last prefill, $\alpha = -5.54$ | 0.23 | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | rejected |
| `think`: addition of the "conclusion" direction | L31, every reasoning token, $\alpha = -0.5$ | 0.25 | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | rejected |
| `value`: push toward low value (Value Axis) | L24, last prefill, $\alpha = 0.5$ | 0.50 | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | rejected |
| `value`: push toward low commitment (Value Axis) | L24, last prefill, $\alpha = 0.5$ | 0.50 | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | rejected |
| `value`: push toward low value (Value Axis) | L18, last prefill, $\alpha = 0.5$ | 0.50 | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | rejected |
| `value`: push toward low commitment (Value Axis) | L18, last prefill, $\alpha = 0.5$ | 0.50 | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | rejected |
| `neurons`: zero the top-16 MLP neurons writing `answer` | L20–31, whole generation | – | 4 | 4/4 [51, 100] | – | – | 0/4 | rejected (bound) |
| `neurons`: zero the top-64 MLP neurons writing `answer` | L20–31, whole generation | – | 4 | 4/4 [51, 100] | – | – | 0/4 | rejected (bound) |
| `circuit`: scale C1 (3 heads + 10 neurons) by $1 - \alpha$ | L19/23/31 heads, L30–31 neurons, $\alpha = 0.5$ | – | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | rejected |
| `circuit`: scale C1 (3 heads + 10 neurons) by $1 - \alpha$ | L19/23/31 heads, L30–31 neurons, $\alpha = 1.0$ | – | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | rejected |
| `circuit`: random control: same-norm Gaussian direction removed | L19/23/31 heads, L30–31 neurons, $\alpha = 0.5$ | – | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | control |
| `perceive`: subtract imminent-completion dir. | L15, last prefill, $\alpha = 0.5$ | 0.09 | 24 | 5/8 [31, 86] | 8/8 | 7/8 | 0/24 | rejected |
| `perceive`: subtract imminent-completion dir. | L15, last prefill, $\alpha = 1$ | 0.17 | 24 | 4/8 [22, 78] | 8/8 | 5/8 | 0/24 | rejected |
| `perceive`: subtract imminent-completion dir. | L15, last prefill, $\alpha = 2$ | 0.34 | 24 | 5/8 [31, 86] | 8/8 | 7/8 | 0/24 | rejected |
| `perceive`: subtract imminent-completion dir. | L15, last prefill, $\alpha = 4$ | 0.68 | 24 | 5/8 [31, 86] | 8/8 | 5/8 | 0/24 | rejected |
| `perceive`: random control, same norm | L15, last prefill, $\alpha = 4$ | 0.68 | 24 | 7/8 [53, 98] | 8/8 | 6/8 | 0/24 | control |
| `verify`: add a "verify" vs. "conclude" direction (CAA) | L18, prefill + 20 first tokens, dose 1 | 1.00 | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | rejected |
| `verify`: add a "verify" vs. "conclude" direction (CAA) | L18, prefill + 20 first tokens, dose 2 | 2.00 | 22 | 8/8 [68, 100] | 8/8 | 6/6 | 0/22 | rejected (bound) |
| `verify`: random control, same norm | L18, prefill + 20 first tokens, dose 1 | 1.00 | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | control |
| `verify`: random control, same norm | L18, prefill + 20 first tokens, dose 2 | 2.00 | 20 | 8/8 [68, 100] | 7/8 | 4/4 | 0/20 | control |
| `mit1`: raise the "Wait" output row for completion openings (weight edit) | `lm_head`, 1 row, 2,560 weights | – | 24 | 6/8 [41, 93] | 8/8 | 8/8 | 0/24 | rejected |
| `mit1`: shorten memory of late linear-attention layers (weight edit) | `A_log`, 9 layers, retention $\times 0.8$ | – | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | rejected |
| `mit1`: lengthen memory of late linear-attention layers (weight edit) | `A_log`, 9 layers, retention $\times 1.25$ | – | 24 | 8/8 [68, 100] | 8/8 | 8/8 | 0/24 | rejected |

**Table 2: Every causal test on the 4B false-success target.** Of the two FS+ contexts of most arms, one is an infeasible task where `answer` is correct, so feasible false successes are represented by a single context and 4 samples per arm; both FS+ contexts of the `perceive` arms are infeasible (Appendix Table 5 gives every count without them). Free generation, $k = 4$, seeds shared across arms, six screening contexts from five tasks (two per stratum). "fs+ answer" must fall (gate: $-25$ points and $-50\%$ relative); "fs− answer" and "healthy valid" must stay within 5 points. "Out." is the number of outputs generated out of 24 planned. "Dose": perturbation norm over the mean residual norm at that layer (nominal; – for neuron and head scalings). Brackets: Wilson 95% interval of the fs+ answer rate, ignoring clustering by task. Every screen is a pilot that can only detect large effects, a fall to 4/8 FS+ answers or fewer. The neuron arms use a different set of six contexts (their own original: 6/8 FS+ answers). *Rejected (bound)*: rejected before completion because the gate had become unreachable. *Control*: random direction, not a candidate. The `perceive` arms use their own screening contexts (reference FS+ 6/8, healthy 6/8). The `verify` dose-2 arm was stopped after 23 outputs when a healthy context looped to the token cap; its counts are from the last snapshot before that output. An L28 arm was not run.

behaviour, though not for the probe. This is an empirical, non-total overlap with a strong confound of position and length, not an impossibility: a low AUROC for a linear probe is not an information bound, and even identical subspaces would not force a selective edit to act globally, which would require proportional covariances. The agent's text carries no more: the best text detector reaches AUROC 0.66 against 0.72 for the step index, and chance within the 18 tasks that end both ways; only the hesitation between samples carries a little (AUROC 0.66).

**Compiling an edit.** Directional ablation at every layer and position equals orthogonalising every writer, the embedding and the vision-merger bias (Proposition 1, Appendix O); a scaled projection compiles exactly, whereas an additive steering vector compiles only as a bias, which Holo-3.1's projections lack. The serving path for a compiled checkpoint is tested on null patches, but no candidate reached it.

### 4.4 Online evaluation

Five arms were nonetheless run online, each paired by task with an original on the same harness and server (Table 3). The three reserve runs of the original, spread over 4.4 points, measure the

| Comparison | tasks | first % | second % | Δ points [95% CI] | better/worse |
|---|---|---|---|---|---|
| *Reserve, noise reference: the original against itself* | | | | | |
| Original, three runs, harness/server (local/pod; GitHub/pod; GitHub/Modal) | 44–45 | 43.2 / 46.7 / 42.2 | | spread 4.4 points | – |
| Original, local vs. GitHub harness (both served by the pod) | 44 | 43.2 | 45.5 | −2.3 [−13.6, +9.1] | 3/4 |
| Original, pod vs. Modal (both on the GitHub harness) | 45 | 46.7 | 42.2 | +4.4 [−6.7, +17.8] | 5/3 |
| *Reserve, arms served by the pod, vs. the pod original on the local harness* | | | | | |
| ExPO $\lambda = 1.25$ | 44 | 40.9 | 43.2 | −2.3 [−13.6, +9.1] | 3/4 |
| `answer` output-row edit (stopped early) | 14 | 21.4 | 35.7 | −14.3 [−50.0, +21.4] | 2/4 |
| Corrective prompt (stopped early) | 20 | 25.0 | 30.0 | −5.0 [−20.0, +10.0] | 1/2 |
| *Reserve, arms served by Modal, vs. the Modal original on the GitHub harness* | | | | | |
| CUA-Lite merge $\beta = 0.5$ | 45 | 48.9 | 42.2 | +6.6 [−6.7, +22.1] | 7/4 |
| Closed-form edit of 24 writers (100%) | 45 | 44.4 | 42.2 | +2.2 [−13.3, +17.8] | 7/6 |
| TEST *split (GitHub harness), vs. the pod original* | | | | | |
| ExPO $\lambda = 1.25$ (had not passed the gate) | 120 | 58.6 | 60.1 | −1.5 [−8.9, +5.7] | 11/12 |
| Corrective prompt (partial run) | 38 | 65.8 | 60.4 | +5.3 [−5.1, +15.9] | 4/1 |
| Original, pod vs. Modal (noise reference) | 120 | 60.1 | 58.6 | +1.5 [−4.5, +7.6] | 9/8 |
| CUA-Lite merge $\beta = 0.5$, vs. the Modal original | 120 | 58.8 | 58.6 | +0.2 [−6.5, +7.5] | 11/10 |
| *Primary verdict, rule fixed in advance: reserve and* TEST*, Modal against Modal* | | | | | |
| CUA-Lite merge $\beta = 0.5$ | 165 | 56.1 | 54.1 | +1.9 [−4.3, +8.0] | 18/14 |

**Table 3: No arm beats the run-to-run difference of the original.** OSWorld, paired by task; scores are mean task scores in %, first run against second. Reserve: `confirm_tasks.json`, never used for construction; each arm is compared with an original on the same harness and the same server, and the first row gives the three reserve runs of the original, the run-to-run noise, and the next two compare the original with itself across harnesses and across servers. The local/pod run lacks one reserve task (`thunderbird/dd84e895`, scored 1 by the GitHub/pod run), so the local-versus-GitHub row pairs 44 tasks, on which the GitHub/pod original scores 20/44 rather than 21/45 on all of them. TEST: ExPO went there without passing the gate; the prompt run is partial. The last row is the primary comparison, with a decision rule fixed before its results were read: a gain only if the lower bound of the CI is above zero. 95% CIs by task bootstrap; better/worse counts tasks where the first run scores higher/lower. The ExPO reserve run is stored as `results_val_patched_eager_m40`, whose model directory names `Holo-3.1-4B-expo125`.

run-to-run noise: the original against itself differs by −2.3 points across harnesses and +4.4 across servers, comparable to every effect, each within a 95% CI that includes zero: ExPO $\lambda = 1.25$ −2.3, the `answer` output-row edit 3/14 against 5/14, the corrective prompt 5/20 against 6/20, and, against the Modal-served original, the CUA-Lite merge +6.6 (CI −6.7 to +22.1) and the closed-form edit +2.2. A gain will be claimed only if the merge, against an original on the same Modal endpoint over the 165 reserve and TEST tasks, has a paired 95% CI above zero, a rule fixed before reading the results: it scores 92.5/165 against 89.3/165 (Δ +1.9, CI −4.3 to +8.0), so by the pre-registered rule (lower 95% bound above zero) no detectable gain, while two servings of the original differ by +1.5 points on TEST. ExPO then went to the TEST split without passing the gate, a decision we report as such: 70.3/120 against 72.1/120 (Δ −1.5, CI −8.9 to +5.7). On the reserve, paired effects below about 17 points go undetected with 80% power.

**An external verifier online.** Feci Guard (Appendix G) shows the final screen to a frozen Claude Sonnet verifier that lists each requirement of the task and refuses the answer, naming what is missing, when one is visibly unmet (at most two refusals per episode). Run once on the 120 TEST tasks under a rule and an analysis committed before its results were read, and paired with the original on the same Modal endpoint, it lowers the false "done" rate from 22.5% to 10.8% (−11.7 points, 95% CI −19.2 to −4.2; two servings of the original differ by less than two points), with no measurable change in score (+1.6, −4.4 to +7.7): it repairs 8 tasks and loses 4, so most avoided false "done" remain failures, now without a success claim. We claim no score gain.

**Imminent completion, not false success.** A direction learned inside successful episodes, final `answer` step versus the step before, beats position (AUROC 0.910 at layer 15 against 0.552) but rises before false and true completions alike (14/14 and 14/14 matched pairs), and subtracting it passes no dose of the gate (Appendix F.4).

## 5 A first negative result: loops in Holo-3.1-0.8B

On MiniWoB++ the 0.8B succeeds on 40.7% of 172 rollouts and loops on 24.5% of failures. Of 40 weight edits along the loop direction, the frozen screening winner (1,953,863 weights) lowers the log-probability of loop actions by 2.31 and that of healthy actions by 8.92 mnat/token, a ratio of

0.26 against 0.31 for a random direction. In free generation, the best steering arm reaches a repeat rate of 81.2% only at a healthy-action rate of 58.3%, and the random direction cannot be told apart from the identity hook (Appendix H). The loop direction is an unselective lever, not a usable one.

## 6 Discussion and limitations

**Why detecting is not causing here.** A direction that encodes "this episode is stuck" or "this episode is ending" can be read reliably while the action is computed from features that do not depend on it; probes in computer-use agents can reach high AUC without detecting the intended property [19], and format-confounded probes can lack a functional role [26]. On the 4B, every piece of evidence points to a belief that the task is done, acted on without checking: the model reads the right date on screen and calls the wrong one "next month", or derives the right month, notes that the dates must change, and drops that requirement because the main one is met. If the same belief closes true completions, removing it would suppress them too; the merge, which lowers true completions more than false ones, behaves that way, and the FS− criterion exists to catch it. The ceiling test suggests why narrow edits fail: the next tool token is assembled by many writers at once, after a thought that has already concluded, and the largest doses we tried damaged generation without moving it.

**Implications for agent developers.** We found no direction whose removal lowers false "done" selectively, in screens able to detect only large effects; within our probes, nothing separated false from true completions beyond the episode's length except the hesitation between samples, and no edit that moved the decision showed an established selectivity on feasible tasks. Our results suggest that a fix must bring in information the model does not use when it stops: an external verifier that checks each requirement of the task against the screen before `answer` is accepted (§4.4, Appendix G), or training that targets false successes, for instance with rewards that penalise episodes ending in a false "done". Weight surgery of the kind tested here has not shown that it can supply it.

**Limitations.** Causal screens use six contexts from five tasks, one of the two FS+ contexts being infeasible, and exclude only large effects; we report "no admissible effect found", not "no effect exists". The ceiling test covers three contexts, one of them infeasible, with confounded donors. Online comparisons exclude only effects above about 17 points. The TEST split was used by 5 earlier online runs, one of them ExPO, which had not passed the gate, and then by the guarded agent, and the confirmation reserve, kept for a committed candidate, by five non-admitted arms. Comparisons are many and uncorrected; no intervention on the model is claimed to work, and none would survive a correction; the one positive result, the guard's lower false "done" rate, is the primary criterion of its own registration, from a single TEST run. The direction analysis is exploratory because its second pass stopped early. Labels are automatic, with thresholds fixed before analysis but not validated by humans. The harness is a third-party reproduction with a broader infeasibility rule than the official adapter, so absolute scores are not comparable to published numbers. The merge donor's training data may overlap OSWorld, and the Claude verifier may have seen public OSWorld tasks. The primary online rule was fixed after the merge's reserve scores had been read, so only its TEST part (+0.2 points, 95% CI −6.5 to +7.5) is strictly prospective, and all our registrations are timestamped entries in our own repository, not a public registry. The semantic audits and "independent" reviews are LLM agents, not validated against human raters, and some protocol decisions were taken by an LLM coordinator. The mechanistic analyses are exploratory, select and estimate on the same few contexts, and use artificial teacher-forced inputs. All results concern one model, one benchmark and one harness. The judge was called through the Claude CLI alias `sonnet` (Claude Sonnet, version not pinned); the Holo4-27B weights are `Hcompany/Holo4-27B` from Hugging Face at revision `4de03cfe`, as recorded in our deployment manifests.

**Broader impact.** Agents that claim success they have not reached mislead users and downstream automation; the verifier we evaluate sends desktop screenshots to a third-party model, a privacy cost; our results argue for external verification rather than trust in an agent's own report, and for caution with interpretability-based repairs whose evaluation can be inflated by the three traps we document. We release no new model.

**Related work.** Appendix J relates this work to steering and directional ablation [2, 24, 33], tool-call steering [4, 36], probing pitfalls [3, 11], weight editing and model extrapolation [22, 44], and evaluation by final state [39].

## References


[1] 1azybug. RealtimeGUI: a decoupled computer-use agent framework with a holo-3.1 OS-World reproduction. GitHub repository, commit a763361, 2026. URL https://github.com/1azybug/RealtimeGUI.

[2] Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction, 2024.

[3] Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances, 2021.

[4] Yuqi Chen, Vincent Siu, Yang Liu, Dawn Song, and Chenguang Wang. Tunable tool-call rates in LLM agents via representation steering, 2026.

[5] Thibault Le Sellier De Chezelles, Maxime Gasse, Alexandre Drouin, Massimo Caccia, Léo Boisvert, Megh Thakkar, et al. The BrowserGym ecosystem for web agent research, 2024.

[6] DaveTseng2019. Agent keeps acting and reports success after screen capture fails (UAC secure desktop). GitHub issue #15, hcompai/holo-desktop-cli, opened August 3, 2026, 2026. URL https://github.com/hcompai/holo-desktop-cli/issues/15.

[7] Chandram Dutta. The termination circuit: how reasoning models stop thinking. LessWrong, July 11, 2026, 2026. URL https://www.lesswrong.com/posts/ajhzc6ktEKyFeJFBS/the-termination-circuit-how-reasoning-models-stop-thinking.

[8] Junfeng Fang, Houcheng Jiang, Kun Wang, Yunshan Ma, Shi Jie, Xiang Wang, Xiangnan He, and Tat-Seng Chua. AlphaEdit: Null-space constrained knowledge editing for language models, 2024. ICLR 2025.

[9] H Company. Holo3.1: Fast & local computer-use agents. Blog post, June 1, 2026, 2026. URL https://hcompany.ai/holo3.1.

[10] Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching, 2024.

[11] John Hewitt and Percy Liang. Designing and interpreting probes with control tasks, 2019.

[12] Zhaoyuan Huang, Tianjie Ju, Pengzhou Cheng, Zheng Wu, Yansi Li, Chuanbiao Song, Jun Lan, Huijia Zhu, Weiqiang Wang, and Zhuosheng Zhang. Do GUI agents know when not to act? enabling conflict-aware termination for multimodal GUI agents, 2026.

[13] Nick Jiang, Isaac Kauvar, and Jack Lindsey. The value axis: Language models encode whether they're on the right track, 2026.

[14] Anton Korznikov, Andrey Galichin, Alexey Dontsov, Oleg Y. Rogov, Ivan Oseledets, and Elena Tutubalina. The rogue scalpel: Activation steering compromises LLM safety, 2025.

[15] Dharshan Kumaran. Reported confidence in LLMs tracks commitment more than correctness, 2026.

[16] Dharshan Kumaran, Viorica Patraucean, Simon Osindero, Petar Veličković, and Nathaniel Daw. How LLMs detect and correct their own errors: The role of internal confidence signals, 2026.

[17] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, et al. Efficient memory management for large language model serving with PagedAttention, 2023.

[18] Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model, 2023.

[19] Yanhang Li, Zhichao Fan, and Zexin Zhuang. When AUC 0.998 is not enough: A candidate evaluation protocol for hidden-state probes of indirect prompt injection in multimodal computer-use agents, 2026.

[20] Evan Zheran Liu, Kelvin Guu, Panupong Pasupat, Tianlin Shi, and Percy Liang. Reinforcement learning on web interfaces using workflow-guided exploration, 2018. ICLR 2018.

[21] Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets, 2023.

[22] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT, 2022.

[23] Aayush Mishra, Daniel Khashabi, and Anqi Liu. Steered LLM activations are non-surjective, 2026.

[24] Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering llama 2 via contrastive activation addition, 2023.

[25] Thilo Reintjes, Sivajeet Chand, Derui Zhu, Sushant Kumar Pandey, and Alexander Pretschner. LegacyWorld: Atomicity-aware evaluation of GUI agents for legacy workflows, 2026.

[26] Subramanyam Sahoo, Vinija Jain, Aman Chadha, and Divya Chaudhary. Linear probes detect task format, not reasoning mode in language model hidden states, 2026.

[27] Tianlin Shi, Andrej Karpathy, Linxi Fan, Jonathan Hernandez, and Percy Liang. World of bits: An open-domain platform for web-based agents. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *PMLR*, pages 3135–3144, 2017.

[28] Patrick Queiroz Da Silva, Hari Sethuraman, Dheeraj Rajagopal, Hannaneh Hajishirzi, and Sachin Kumar. Steering off course: Reliability challenges in steering language models, 2025.

[29] Yuan Sui, Yulin Chen, Yibo Li, Xue Jiang, Yufei He, Yihong Dong, et al. TACT: Mitigating overthinking and overacting in coding agents via activation steering, 2026.

[30] Chung-En Sun, Linbo Liu, Ge Yan, Zimo Wang, and Tsui-Wei Weng. LLM agents already know when to call tools – even without reasoning, 2026.

[31] Rikiya Takehi, Jayden Teoh, Alex Zhang, and Omar Khattab. Sparks of weight-space programming: LLMs can natively read & write neural network weights. Blog post, MIT CSAIL, November 2, 2026, 2026. URL https://rikiyat.github.io/blog/weight-space-programming/.

[32] Daniel Tan, David Chanin, Aengus Lynch, Dimitrios Kanoulas, Brooks Paige, Adrià Garriga-Alonso, and Robert Kirk. Analyzing the generalization and reliability of steering vectors, 2024.

[33] Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering, 2023.

[34] Constantin Venhoff, Iván Arcuschin, Philip Torr, Arthur Conmy, and Neel Nanda. Understanding reasoning in thinking language models via steering vectors, 2025.

[35] Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small, 2022.

[36] Zekun Wu, Ze Wang, Seonglae Cho, Yufei Yang, Adriano Koshiyama, Sahan Bulathwela, and Maria Perez-Ortiz. Tool calling is linearly readable and steerable in language models, 2026.

[37] Zhengxuan Wu, Aryaman Arora, Atticus Geiger, Zheng Wang, Jing Huang, Dan Jurafsky, Christopher D. Manning, and Christopher Potts. AxBench: Steering LLMs? even simple baselines outperform sparse autoencoders, 2025.

[38] Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming Xiong, Victor Zhong, and Tao Yu. OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments, 2024.

[39] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. $\tau$-bench: A benchmark for tool-agent-user interaction in real-world domains, 2024.

[40] Mohammad Zbeeb, Hasan Abed Al Kader Hammoud, and Bernard Ghanem. Reasoning vectors: Transferring chain-of-thought capabilities via task arithmetic, 2025.

[41] Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods, 2023.

[42] Chengfeng Zhao, Yuqiao Tan, Shizhu He, Yequan Wang, Jun Zhao, and Kang Liu. Neural procedural memory: Empowering LLM agents with implicit activation steering, 2026.

[43] Tianyi Zhao, Yinhan He, Wendy Zheng, Yujie Zhang, and Chen Chen. Wired for overconfidence: A mechanistic perspective on inflated verbalized confidence in LLMs, 2026.

[44] Chujie Zheng, Ziqi Wang, Heng Ji, Minlie Huang, and Nanyun Peng. Model extrapolation expedites alignment, 2024.

[45] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, et al. Representation engineering: A top-down approach to AI transparency, 2023.

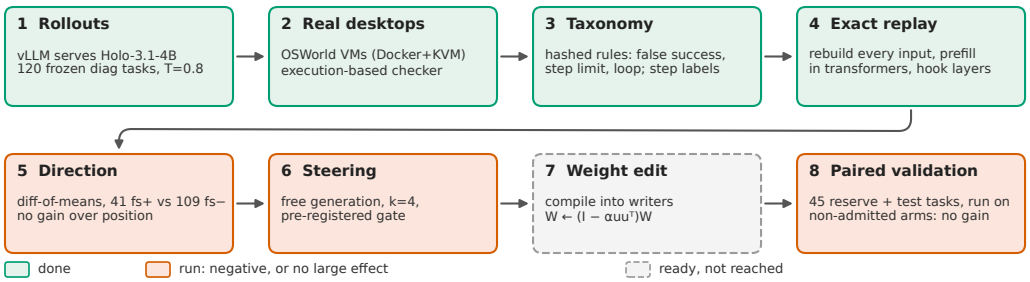



**Figure 5: The pipeline and where it stands.** Stages 1–4 are complete and verified. The official direction analysis (5), on 41 admissible FS+ steps, finds no signal beyond position. Steering (6) has been run, and no candidate shows a large effect. The compiled weight edit (7) is built and tested on null patches, but not reached because no candidate has been admitted. The paired online validation (8) was run anyway, on arms that had not been admitted (§F.6), and shows no gain beyond the run-to-run noise of the original.

# A About the appendices

The main text condenses the analyses below. Appendices B–L give each of them in full, with every number, the remaining figures and tables, the compilation of interventions into weights (Proposition 1) and the 0.8B study; figures and tables already shown in the main text are referenced rather than repeated. Figure 5 summarises the pipeline and where it stands.

# B Setting

**Models.** Holo-3.1 models are fine-tunes of the Qwen family [9]; their released configuration is a Qwen3.5 vision-language model.

**OSWorld harness (4B).** We reuse the Holo agent loop as packaged by RealtimeGUI at commit `a763361` [1], a third-party reproduction rather than an H Company repository, changing only the model provider. Its rule for recognising a declared infeasibility (`task.{0,4}infeasible|infeasible|cannot be (completed|done)|not possible`) is broader than that of the official OSWorld adapter for Holo, which accepts only `task.{0,4}infeasible`; our absolute scores are therefore not strictly comparable with H Company's, while paired comparisons within our harness are. vLLM 0.23 serves the 4B with thinking enabled, $T = 0.8$, at most 50 steps and the three latest screenshots in context. OSWorld [38] tasks run in Ubuntu VMs under Docker+KVM, at most four in parallel, and are scored by the benchmark's execution-based checkers. We fixed three infrastructure issues in the VM provider (a memory cap, binding VM ports to localhost, an nftables network script) and modified no task or checker.

**Frozen split.** From the 361 tasks of RealtimeGUI's `test_nogdrive.json`, we sort ids by SHA-1 within each domain and deal them alternately into a DIAG and a TEST pool, keeping 120 tasks each with per-domain quotas. The split files have SHA-1 prefixes `51e670b3` (diag) and `957d68e7` (test). No construction or screening read the test split. 5 online runs then used it (§F.6): the original, served once by the pod and once by Modal, the ExPO edit, which went to it without having passed the gate, a partial corrective-prompt run, and the CUA-Lite merge under a decision rule fixed in advance.

**MiniWoB++ harness (0.8B).** Our first attempt used Holo-3.1-0.8B on MiniWoB++ [20, 27] through BrowserGym [5], with the task area cropped and upscaled $5\times$; it is described in §H and Appendix R.

**Hardware and licences.** Rollouts, replays and screens ran on one NVIDIA H200 shared between the vLLM server and research jobs; edited checkpoints were also served on rented H100 GPUs, and the OSWorld VMs ran on a separate CPU host. As order-of-magnitude estimates, the project used about 14 hours of one H200 (rented, shared by every research job and the rollout server, so per-experiment GPU-hours are not separable), about 8 H100 GPU-hours (USD 31.9 billed over two accounts) and about 60 GitHub Actions CPU jobs for the OSWorld VMs. Holo-3.1-0.8B and -4B,

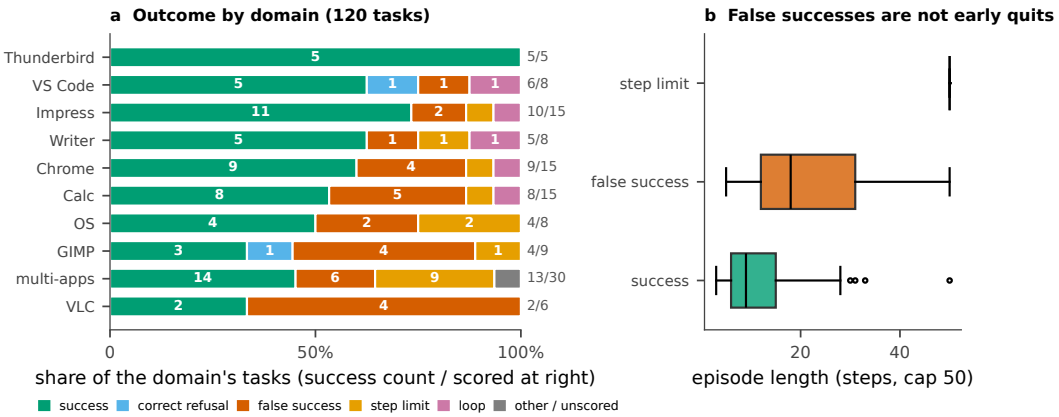



**Figure 6: False success is spread across domains and is not an early quit.** Holo-3.1-4B, OSWorld diagnostic split, 50 steps, $T = 0.8$. (a) Primary outcome per domain, sorted by success rate; numbers in bars are task counts, and the right column gives fully solved tasks over scored tasks (the *success* category also counts the 2 partially scored tasks). (b) Episode length by outcome. The full table is Appendix Table 8.

Qwen3.5-4B, the CUA-Lite donor, OSWorld and BrowserGym are released under Apache-2.0 and MiniWoB++ under MIT; the RealtimeGUI repository has no licence file at the cited commit, and we use it unmodified except for the model provider.

# C Most failures are a false "done"

**Taxonomy.** Each episode gets one primary category from fixed rules over its actions, screenshots and final score. *False success*: the agent calls `answer` on a feasible task scored 0, or claims completion on an infeasible task. *Loop*: at least three consecutive identical actions while less than 1% of pixels change. *Step limit*: the 50-step budget is exhausted without a loop. At step level, FS+ marks the final `answer` of a false success and FS− that of a true one. A correctly worded refusal of an infeasible task is excluded from both. The rule file is hashed before any direction is fitted (current hash `1e5079afb500`); later changes are logged with their hash and none altered a label. Before fitting any direction we also require that the target category cover at least 25% of failures and provide at least 40 positive steps (the GO rule).

**Result.** On the diagnostic split (120 tasks, 119 scored), Holo-3.1-4B scores 56.7% (the official OSWorld score, partial credit included, the convention used for every score in this paper) and fully solves 66 tasks: 55.5%, 95% Wilson CI 46.5–64.1. An independent reproduction with the same public harness reports about 45.5% at 100 steps on a different task set [1]; we do not claim the two numbers are comparable. Of the 51 failures (Figure 6), 29 (57%) are false successes, 16 (31%) have the step limit as primary category (21, or 41%, exhaust the 50 steps at all, loops included), 5 (10%) loop, and 1 is unclassified; 1 further task is unscored (the "Other + unsc." column of Table 8 counts both). 9 false successes are claims of completion on infeasible tasks, and 20 are feasible tasks done wrongly or partially. On an infeasible task the correct action is `answer` with a refusal, so $P(\texttt{answer})$ is the wrong metric there; one such context, Chrome `480bcfea` at step 4, enters several screens below, and Appendix M gives every count with and without it. These are mostly not misworded refusals: among the 75 tasks open to construction, 6 are infeasible (11 episodes), and of the 10 final answers 8 claim success, while only 1 task states its inability, in words the harness rule misses. They are not early quits: the median false success ends after 18 steps. Every domain has false successes except Thunderbird, and Chrome, Calc, GIMP and VLC have four or more each (Fig. 6a). False success passes the share criterion of the GO rule, but its 27 FS+ steps alone do not pass the count criterion. With a second pass over the same tasks, stopped at 80 scored tasks when the GPU returned to production, 41 FS+ and 109 FS− steps are admissible, above the required 40 (as of 2026-11-07 21:56:45 UTC).

### C.1 The model says "done" on most false completions too

We re-sampled the original model on six screening contexts from five tasks, two per stratum ($k = 4$, production sampling, thinking enabled, invalid outputs counted; Fig. 1c and Table 2, first row). It calls `answer` on 8/8 false-completion samples and 8/8 true ones, and never on mid-episode contexts (0/8). It reproduces the terminal versus non-terminal distinction exactly, and on these contexts shows no false versus true distinction; on the larger, strictly feasible cohort of §F.5, it answers on 72% of false and 94% of true completions. Any repair must therefore change a decision the model takes with near certainty in both cases, and only in one of them.

The reasoning has the same shape in both cases. On 16 re-sampled reasoning traces at completion steps of 4 construction tasks, 8 on false and 8 on true completions, every trace opens with a success claim ("The task has been completed successfully", "The document has been saved successfully") at generated-token index 4–5, counted from zero, with no verification word before it (0/16), then lists the steps taken (median 116 tokens). Because true completions show it too, this is a property of Holo's output format at completion steps, not a marker of false success. It locates the written conclusion, not the moment the decision forms inside the model. A direction fitted on these tokens separates the two pairs perfectly, but so do context length and a single lexical cue, so we give it no weight; steering along it is negative (Table 2). What the format skips is the check: the Spotify episode states that the installation cannot finish and claims success one step later (Fig. 1a).

## D From a served agent to activations

Rollouts are served by vLLM [17], which does not expose activations. We therefore never record activations during rollouts. For each recorded step we *rebuild* the exact chat the agent sent: system prompt, task, `<observation>` blocks, at most three screenshots with older ones replaced by `[screenshot evicted]`, parsed tool calls and tool outputs. We render it with the model's chat template and run one forward pass in `transformers`, hooking the residual stream and MLP output of every layer at two positions: the last prefill token, and the token before the action JSON.

**Fidelity is tested, not assumed.** (a) *Messages.* Running the real agent loop with a fake client and environment, the rebuilt messages are identical to those sent on 7/7 steps, also when three consecutive parse failures are injected before steps 1, 4 or 5, the harness's image-eviction edge case. (b) *Tokens.* The prefill length in `transformers` equals vLLM's `prompt_tokens` on 12/12 steps of a real trajectory (4,459 to 10,432 tokens), and the replay loop repeats this check on the first and last step of every task. (c) *Likelihood.* The replayed model gives its own recorded actions a mean log-probability of $-0.16$ per token, against $-1.10$ for synthetic actions. One divergence is invisible in the files: an exception in `env.step` leaves an unrecorded assistant message in the live context. The diagnostic pass logged 0 such events and 6 triple parse failures, which test (a) shows to be harmless.

## E Three traps that would each have produced a positive result

**Trap 1: the direction measures position.** False completions come later in an episode than true ones, so anything that grows with the episode separates them. A first difference-of-means direction between FS+ and FS− steps (22 and 55 steps), cross-validated by task, peaked at AUROC 0.836 against 0.870 for the step index and 0.880 for context length. Reported without baselines, 0.836 reads as a solid detector; in fact it was worse than counting steps. The official analysis, run once FS+ steps passed the pre-registered 40 (41 FS+, 109 FS−, 95 tasks, both passes; exploratory because the second pass stopped early), confirms it (Table 1, Fig. 2b). The best direction, at layer 14 on the last prefill token, reaches 0.76 (logistic probe 0.71) against 0.724 and 0.735 for the two position baselines, and the token before the action does worse (0.70). Controlling for position directly, a logistic model on step and log context length scores 0.722; adding the best direction raises it by at most 0.04, at a layer chosen after the fact, within noise. We find no false-success signal in the residual stream beyond the length of the episode, which is consistent with the failure of every direction-based intervention below. A value-axis direction in the style of Jiang et al. [13] shows the same confound in a separate construction (appendix Q). Our protocol therefore matches positives and negatives on step and context length before fitting, and reports both baselines next to every AUROC.

**Trap 2: regression to the mean.** Agent outcomes are stochastic at $T = 0.8$. Selecting tasks because they failed in one run selects partly for bad luck. Re-running the unmodified agent on the same tasks, only 9/18 (50%, 95% CI 29–71) pass-1 false successes are false successes again (an earlier snapshot of the second pass, on fewer tasks, gave 42%), and 6 of them now succeed (Fig. 2c). A patch evaluated as "false-success tasks before versus after" would therefore be credited with about 50 points of reduction while doing nothing. Our validation compares two fresh runs, original and patched, on a frozen reserve of `confirm_tasks.json` tasks that no intervention was built on (§E.1).

**Trap 3: detecting is not causing.** The 0.8B loop direction is a textbook detection result (Fig. 2a): task-grouped AUROC 0.893 at layer 10, against 0.70 for the step index and 0.72 for context length, replicated at 0.904 on an independent refit. It is a poor lever in every causal test we ran: it reduces loops only at a large cost to healthy actions (§H). Projecting it out of every writer erases the marker from the residual stream (the loop/non-loop projection gap at layer 10 falls from 0.1653 to $-0.00008$) and leaves loop actions as likely as before relative to healthy ones. The marker can be deleted without touching the behaviour it marks.

### E.1 The protocol these traps impose

1. **Free generation is the only efficacy measure.** Teacher-forced log-probabilities of a recorded action mix format, arguments and end-of-sequence, and on the 4B the decision is taken during reasoning, after which $P(\texttt{answer})$ is close to 1 for true and false completions alike. Each candidate regenerates the full reasoning ($k = 4$ per context, fixed seeds shared across arms, production sampling with `top_k=20`), and invalid or truncated outputs stay in every denominator.
2. **A gate fixed before any data.** On FS+ contexts of tasks not used to build the intervention, $P(\texttt{answer})$ must fall by at least 25 points and 50% relative. The FS− answer rate and the rate of valid non-answer actions on mid-episode contexts may each fall by at most 5 points, and invalid outputs may rise by at most 5 points. The FS− criterion is what separates "less false success" from "less success".
3. **Controls.** A random direction with the same norm and scope, an identity hook that measures the noise floor, and a fixed budget of generation batches per hypothesis, counted before execution, failures included. Two consecutive settings without progress stop a hypothesis.
4. **Disjoint confirmation.** The official direction analysis is gated at 40 admissible FS+ steps. The `confirm_tasks.json` reserve is excluded from every construction and screening, and was to be read only to confirm a committed candidate, with both arms re-run fresh. In the event no candidate was admitted, and five non-admitted arms were run on it (§F.6); we list this as a limitation (§K).

## F No selective effect found

### F.1 Where the tool choice is carried

Before steering, we mapped where the choice of `answer` is made. On two FS+ and two FS− completion steps, we patched the residual stream from the same task's previous step (whose recorded action is a click) at each of 32 layers and four positions, 512 patches in all, with the original reasoning held fixed. The quantity is the logit margin between the first token of `answer` and that of the best other tool. Identity patches give exactly zero.

The decision is invisible at the end of the prefill and barely visible mid-reasoning: the largest mean margin change is -0.19 and -2.28 logits there, for FS+. It appears only at the tool-name position, from about layer 16 upward, where patching moves the margin by up to -34.8 logits on FS+ and -37.5 on FS− (Fig. 3). Component scans single out three attention heads, L19/H15, L23/H1 and L31/H14, and ten late MLP neurons. Their attention falls mostly on the system-prompt definition of `answer` (L19/H15, L23/H1) and on the JSON being written (L31/H14). Removing all of them together lowers the margin by 11.2–12.2 logits, and flips 0 of the 4 decisions (smallest remaining margin 2.9). Across all 992 component patches and 28 joint patches, 0 and 0 decisions flip: the components move the margin, not the action. The only flips come from residual patches (21 on FS+ and 23 on FS− of 512), at the tool name in the last layers, where replacing the residual stream transplants the donor's choice into the output head. They show where the choice is carried, not

where a false judgement is formed. The effect is the same on true and false completions (Fig. 3c). It is a generic tool-retrieval circuit, not a false-success circuit.

**A ceiling test.** The patches above replace one layer at a time; a ceiling test asks how much must be replaced at once. With the recorded input forced and only logits read, the outputs of whole blocks of writers are swapped in at the decision position from a donor step where the model clicked: the same task's previous step for two FS+ contexts, another task's step for one true completion (Fig. 4). On the first FS+ context (margin +16.9), all 32 MLPs together bring the margin to +0.9 without flipping it, while the MLPs and token mixers (attention and DeltaNet) of layers 12–23 flip it ($-3.5$), as do those of layers 24–31 ($-10.0$), and those of layers 0–11 do not (+16.1). The second FS+ context behaves the same (MLPs +2.1, block 12–23 $-3.0$), and so does the true completion (block 12–23: from +16.4 to $-2.2$). Swapping with itself and swapping the final residual reproduce the baseline and the donor exactly. Three caveats bound this result. The first FS+ context is an infeasible task, on which `answer` (with a refusal) is the correct tool (Appendix M). The donor differs in screen, reasoning, thought and position, not only in decision. And since a true completion flips too, the test locates where the identity of the next tool token is assembled, not a mechanism of false success. No particular block is needed: the MLPs and mixers of layers 12–23 or of layers 24–31 each flip the first context, and removing a single direction of the swap difference from the writers of layers 12–23 brings the margins of the two FS+ contexts to +1.4 and +1.6, but also those of two true completions to +4.8 and +2.8. It is consistent with the failure of every edit confined to one or two layers or to the MLPs, and it says nothing about which edit would separate false from true completions.

### F.2 The tool call leans on the text already written

The tool call leans on the text the model has already written, for true completions as for false ones. On the 19 false and 20 true completions of the strictly feasible cohort (§F.5), teacher-forced with the screen, history, reasoning and token count fixed, replacing the note and thought by a neutral text of the same token length ("Observing the current screen.") removes 14.0 logits of `answer` margin on false completions (95% CI 12.9–15.1; original 17.3) and 14.5 on true ones (12.9–16.0; original 18.0), 82% and 80% of the margin per task. The two losses differ by $-0.46$ logits ($-2.36$ to +1.50; label permutation $p = 0.65$). A counterfactual text of the same length ("The task is not complete yet; I still need to inspect the screen.") leaves 0.10 and 0.61 logits, nearly cancelling the margin on average (`answer` still leads in 10/19 and 13/20 contexts), while neutralising the recorded reasoning instead removes only 4.1 and 4.8. One true completion has an empty note; without it, the true-completion loss is 15.0. The written note and thought thus carry 80–82% of the `answer` margin in both strata, yet without them `answer` remains the preferred tool in 37 of 39 contexts (18/19 false, 19/20 true): the conclusion is reinforced by the note, not created by it, although the success claim in the retained reasoning may carry the rest. This is an attenuation of the margin, not a change of decision. Nor does a probe find a robust discriminator in the residual stream of these contexts: of 72 out-of-fold probes (6 layers, three positions, four models), the best, chosen after the fact, reaches AUROC 0.64 (0.47–0.80) against 0.70 for the raw step index, and its permutation $p$ of 0.04 becomes 1.0 after correction for the 72 probes. With these sample sizes a pre-specified score would need an AUROC of at least 0.725 (one-sided) to 0.755 (two-sided) to be detected with 80% power, so the absence of the information is not proven. These are artificial inputs, and the joint replacement does not separate the wording of the conclusion from the details the note describes.

A first version of the test, on two FS+ contexts, illustrates it with a donor's text rather than a neutral one: it keeps the screen, the history and the weights, and replaces only the text the model has generated at the decision step with that of the same task's previous step, where the model clicked. On the first FS+ context (margin +16.9), the donor's reasoning alone lowers the margin to +7.6, the donor's JSON prefix alone (note and thought) to +1.1, and both together flip it ($-7.4$). The second context behaves the same (+18.4; reasoning +11.1, JSON prefix $-3.1$, both $-7.6$). Three limits weighed on that version, the first of which the cohort test above removes: it had no control on true completions or with a neutral text. Only its second context is feasible, and the donor is the previous step, whose text announced a click, so that copying its text largely copies its decision. And the text does not account for everything: with all of it swapped, the margin stays 9.8 and 14.5 logits above the donor's, a share that comes from the screen and the history. Within these limits, the tool call depends causally on the text already written, the note and thought weighing most, and the 13.9–19.1-logit margin of §F.3 is largely conditioned on that text rather than an intrinsic robustness of

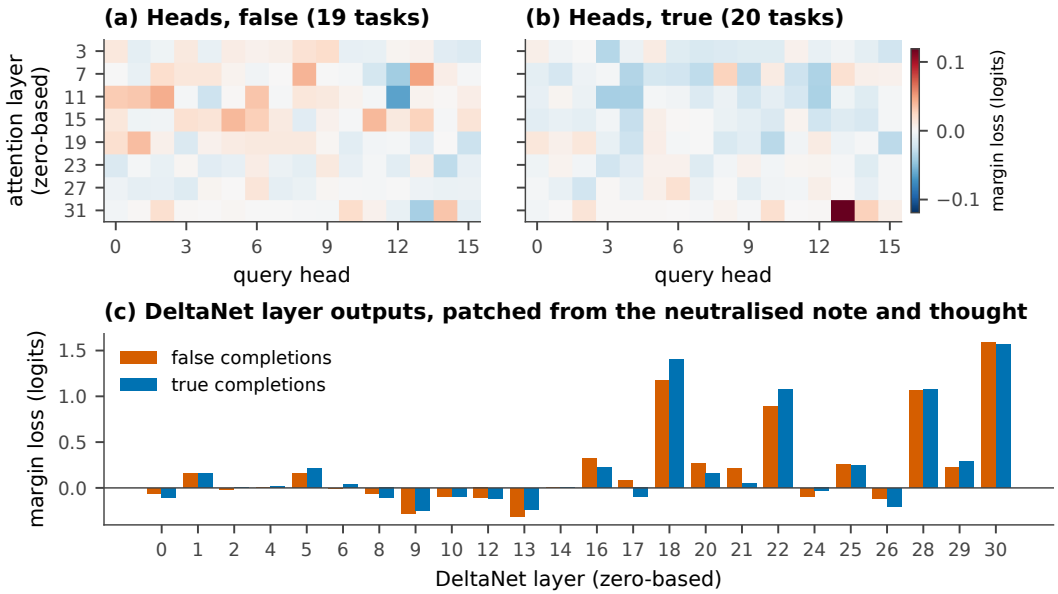



**Figure 7: The written conclusion reaches the tool choice through the DeltaNet memory, for true completions as for false ones.** Teacher-forced `answer` margin lost when the note and thought are neutralised, on the strictly feasible cohort. (a, b) Blocking the final query's attention to the note and thought, one head at a time: at most 0.12 logits (colour scale). (c) Replacing one DeltaNet layer's output at the decision position by its value under the neutralised note and thought: 0.9–1.6 logits at layers 18, 22, 28 and 30, alike for both strata (note the ten times larger scale). Single-layer patches do not add up to the whole effect (7.55 and 8.18 logits). Heads and layers are screened on the same cohort: exploratory.

the decision. An attribution on the same cohort locates the channel. Replacing the inner tokens of the note and thought with spaces, at the same length and positions and with the screen, history and reasoning kept, lowers the teacher-forced margin by 7.55 logits on false completions (mean 17.3 before) and by 8.18 on true ones (mean 18.1 before, measured on its own cached replay path), less than the neutral text above, and lowers it on every context of both (19/19 and 20/20); the difference between the two losses is $-0.64$ logits (95% CI $-2.13$ to +0.80). The note and thought carry the decision as much for true as for false completions: the channel is real, but not selective. It runs chiefly through the recurrent memory rather than attention (Fig. 7). Replacing the output of a single DeltaNet layer at the decision position by its value under the neutralised text costs 0.9–1.6 logits at layers 18, 22, 28 and 30 (zero-based) and at most 0.33 at the other layers. Blocking the final query's attention to these tokens, one head at a time over the 128 heads of the eight full-attention layers, costs at most 0.048 logits on false completions and 0.12 on true ones, and at best 0.05 logits more on false than on true. In this hybrid model, the conclusion already written is carried forward by the recurrent memory. The same contexts define the comparison and the cohort, so this screen is exploratory, and layer-output patches locate carriers of the change, not its mechanism.

### F.3 Every steering arm

Table 2 lists every intervention run on the 4B false-success target, each under the gate of §E.1 and on the same six screening contexts. Each line corresponds to a distinct hypothesis taken from the literature.

- *Tool choice is linearly steerable* [36]: add the position-matched FS+−FS− direction at the last prefill token, at layer 16 ($\alpha = -1.20$ times its norm) and layer 24 ($\alpha = -5.54$).
- *The decision is taken during reasoning* (§C.1): add a "conclusion" direction, fitted on the reasoning tokens, at every reasoning token of layer 31.
- *False success is excess expected value* [13, 15]: push toward low value or low commitment at layer 24 or 18.
- *Late MLP components write the decision* [7], tested here at the neuron level: zero the top-16 or top-64 neurons by direct attribution to the `answer` logit.

- *The model fails to check*: add a "verify" versus "conclude" direction (contrastive activation addition) at layer 18 at 1 and 2 times the residual norm, over the prefill and the first 20 generated tokens.
- *False success is a perception of completion*: subtract the imminent-completion direction of §F.4 at layer 15.
- *The model opens with its conclusion* (§C.1): raise the output row of the token "Wait" for the states that open a completion, a 2560-weight edit of the output head.
- *Memory of the early episode fades*: lengthen or shorten the retention of the nine late linear-attention layers by a factor 1.25.
- *The circuit of §F.1 carries it*: scale its three heads and ten neurons (C1) by $1 - \alpha$ with $\alpha = 0.5$ and 1, against a random direction of the same contribution norm.

No arm passes the gate, and none shows a large effect. Most leave the false-"done" rate at its original value, including removing the whole C1 circuit ($\alpha = 1$), which answers on every false completion, like the random control. The neuron arms, screened on six other disjoint contexts whose own baseline is 6/8 FS+ answers, were stopped by a mathematical bound: with 4/4 answers already observed on the first unseen FS+ task, at most 3 answers out of 8 would have been admissible. The value-axis arm did change the text. An independent reader found an explicit intention to re-evaluate in 3/8 FS+ reasoning traces against 1/8 at baseline, but 0 verification actions and the same answer rate. The model wrote that it should check, then answered anyway.

The "Wait" row is the one edit that lowers false answers without touching anything else: 6/8 on false completions, with 8/8 true completions kept, 8/8 healthy actions valid and 3/3 general-text controls token-identical. It still fails the 50% relative criterion, and an independent review rates none of the two replacement actions useful: one reloads a settings page for an invented reason, the other scrolls without fixing the dates. Making the model pause is not enough to make it repair. A conditional variant adds an early end-of-reasoning row to the same edit (5101 weights). Inside its construction set, 5/8 false completions still answer. On one Budget sample the model reasons correctly that next month is December, says the dates must be changed, and generates a click on the date summary of the results page; it was not executed. These are intentions to correct, not corrections: an independent review of the screenshot finds that the dates sit in plain summary text, next to a separate "Modify Rental Details" control, and that no step proves the summary opens an editor. All 7 such generated clicks, over 4 lots of this edit and its diagnostics, are reclassified as unverified, 0 confirmed useful; no effect on the interface or on the OSWorld score is shown. Nor does the edit transfer. On nine training tasks left out of its construction, the final audit counts 0/12 useful actions on false completions (original 0/12), 3/12 outputs that name the real error (original 1/12), and 7/12 false finishes (original 8/12), while true completions go from 10/12 to 9/12. The edit leaves a mechanistic signal, the model recognising the error then rationalising it away, and no demonstrated useful effect.

**These screens are pilots.** Each efficacy cell holds 8 generations from two tasks. 8/8 has a Wilson 95% interval of [68, 100]%, and samples clustered in two tasks make the real uncertainty larger, so a rejection means only that no large effect appears on these contexts, not that none exists. By design the gate flags a reduction to at most 4/8 answers; smaller effects pass unnoticed. Table 2 gives the interval of every efficacy cell.

**A ceiling effect is likely.** At the tool-name position the margin of `answer` over the best other tool is 13.9–19.1 logits on false completions and 10.9–17.8 on true ones. This position comes after the JSON `thought` field, which the output schema writes before the tool call and which already states that the task is complete, so the margin measures how confidently the model carries forward a decision it has already verbalised, not the margin of the decision itself. Even so, the perturbations tested are small: next to the residual stream, they are 0.13 (layer 16) and 0.23 (layer 24) of the mean residual norm for the matched direction, 0.25 for the reasoning-token direction (difference-of-means norm 24.5 against a residual norm of 50.0 at layer 31), and 0.5 for the value axes, while removing the whole C1 circuit lowers the margin by 11.2–12.2 logits without flipping it. Larger doses do not break the ceiling either. The "verify" direction at one and two times the residual norm leaves FS+ at 8/8; at dose 2 a healthy context loops on "I see" up to the 3,072-token cap, which stopped the arm after 23 of 24 outputs. Dose sweeps at layers 24 and 31 with random controls were not run.

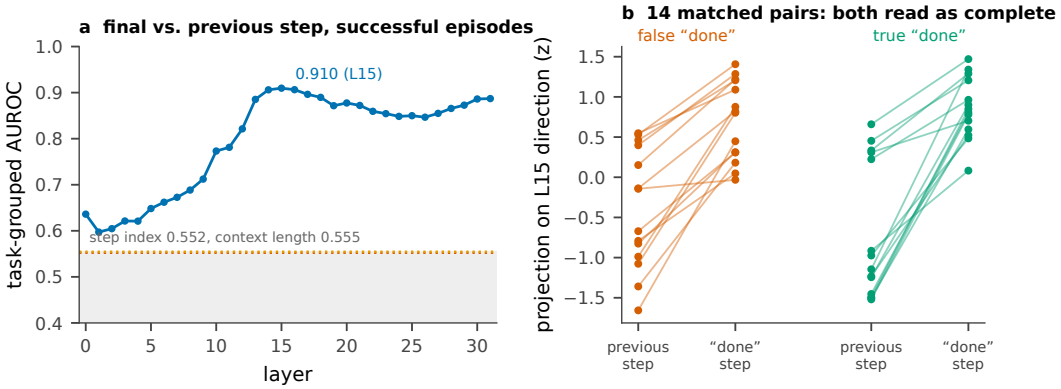



**Figure 8: An imminent-completion direction beats position but does not separate false from true completions.** Exploratory, no causal test. (a) Task-grouped AUROC of a direction learned inside successful episodes (final `answer` step vs. the step before it, 58 pairs, 40 tasks), by layer; the grey band is at or below the position baselines. (b) Cross-fitted projections at layer 15 on 14 matched pairs: the projection rises at the “done” step for false and true completions alike.

A second target, getting stuck at the step limit, shows the opposite failure of steering. A residual addition at layer 16 or 24 damaged healthy actions before any stuck context was generated; after 5/24 outputs, the best reachable health rate was 91.7%, below the required 95%.

### F.4 A detector of imminent completion beats position, not false success

All directions above contrast false and true completions, which differ in position. `astra-perceive` sidesteps this by learning a direction *inside successful episodes*: the last-prefill residual of the final, correct `answer` step minus that of the step just before it, on 58 pairs from 40 construction tasks (reserve excluded, cross-validation grouped by task across both passes). Because both members of a pair sit next to each other in the same episode, position cannot separate them across tasks. The direction peaks at layer 15 with task-grouped AUROC 0.910, against 0.552 for the step index and 0.555 for context length (Fig. 8a). This is the first signal in this study that position does not explain, in contrast to the false-success direction of Table 1. Within a pair, both baselines are trivially perfect, so the pooled AUROC is the relevant comparison, and the layer was chosen on the same cross-validation.

Read on 14 false/true completion pairs matched on pass, step and context length, with each task’s direction fitted without that task, the direction rises at the “done” step in 14/14 false and 14/14 true completions (mean 1.11 and 1.54 standard deviations; Fig. 8b). At the terminal step, false completions sit -0.14 standard deviations from true ones, above them in only 6/14 pairs. It is therefore a detector of imminent completion, not of false success: it fires before a false “done” as it does before a true one. On these 14 pairs, the projection reads a false completion as complete. This is compatible with a perception error: the screen is taken as showing a finished task. Two same-task image pairs from construction data illustrate it, a spreadsheet cell claimed green whose text has no green pixels, and rental dates claimed for next month while the page shows the current one. It does not separate misperception from commitment to the action. Subtracting it at layer 15 (up to 0.68 of the residual norm, on its own screening contexts whose reference is 6/8 FS+ answers) gives 5/4/5/5 FS+ answers out of 8 for $\alpha = 0.5, 1, 2, 4$, against 7/8 for a random direction of the same norm, and no dose passes the gate (Table 2). The few drops are not repairs: the non-answer actions that replace `answer` are empty or invented text entries, such as a made-up e-mail address. Attention maps on two same-task pairs found no head, out of 80 tested, that reads the evidence region more when the model succeeds (0/80 pass the frozen rule). The perception hypothesis is not refuted by this small screen, but it has no support beyond the projection of Fig. 8.

### F.5 Weight edits: levers without established selectivity

Weight edits that need no direction, or that target a mechanism, are detailed in Appendix N; we summarise them here. Extrapolating the fine-tune away from Qwen, $W(\lambda) = W_{\text{Qwen}} + \lambda\,(W_{\text{Holo}} -$

$W_{\text{Qwen}}$), called ExPO [44], leaves false answers at 16/16 in every arm but one (15/16, MLPs only, where true completions drop more); the edit at $\lambda = 1.25$ keeps text and JSON intact and runs online (§F.6). The mechanism-level edits H2–H10 (memory gates, MLPs or all language tensors moved toward Qwen, the end-of-thinking embedding, the attention temperature, the rotary share of the query gain) leave the tool call in place, break generation, or move the call only on the infeasible context, where theme clicks replace an answer that should have explained the impossibility (Table 7); two-neuron circuits, a narrow MLP edit adapted from weight-space programming and a corrective instruction injected into the reasoning do no better. Two edits do move the decision on feasible contexts, and neither shows an established selectivity.

Merging in a donor trained on desktop tasks moves the decision globally, not selectively. Adding a fraction $\beta$ of the task vector of CUA-Lite, a Qwen3.5-4B fine-tune on ScaleCUA desktop data, to the language layers of Holo ($W_{\text{Holo}} + \beta\,(W_{\text{CUA}} - W_{\text{Qwen}})$; vision tower unchanged; input embeddings merged, the output head being tied to them at load time), and serving it with the same 51 calls as the exact original model, re-served on the same GPU type and vLLM version (in a separate account and container), lowers false-completion answers to 15/16 at $\beta = 0.25$ and 11/16 at $\beta = 0.5$, and true-completion answers further, to 11/16 and 2/16 (original 16/16 and 16/16), with every healthy action valid and no invalid output; on the feasible FS+ context alone, $\beta = 0.5$ answers 3/8. The dose–response is a global drop of `answer`, not a discrimination. An independent audit finds the replacing actions coherent and unnecessary: scrolls after the model has recognised success, and verification clicks after true completions. The model continues instead of concluding, whatever the truth of the completion. A donor trained on filtered web tasks, Fara1.5-4B, leaves the decision in place at $\beta = 0.5$ (16/16 and 14/16). BF16 rounding blurs small merges: at $\beta = 0.5$, 88 of the 426 language tensors are left unchanged and the update norm is 2.34 against an intended 2.01; at $\beta = 0.25$, 101 tensors are unchanged. CUA-Lite's training data may overlap OSWorld tasks. Online, the merge shows no detectable gain over an original served by the same endpoint: +1.9 points over 165 tasks (95% CI −4.3 to +8.0; §F.6).

A closed-form edit of the writers that the ceiling test points to moves the decision as well, unselectively on the screening contexts; on a larger feasible cohort its selectivity is not established either way (below). In the spirit of AlphaEdit [8], the 24 residual writers of layers 12–23 (MLP down-projections and mixer output projections) receive a ridge update that maps FS+ keys (constructed on the ceiling-test contexts, the first of them infeasible) toward the donor displacement while being projected off the span of protected keys (true completions, healthy steps and text prompts). On contexts outside its construction, served next to the original on the same GPU, false-completion answers fall from 15/16 to 8/16 but true-completion answers fall further, from 15/16 to 3/16, with 16/16 healthy actions valid and 0 invalid outputs. The whole FS+ drop comes from one of the two contexts (7/8 to 0/8; the other stays at 8/8). A first visual audit of the replacing actions, which had described the screenshot of the previous step, was retracted; the corrected audit finds that on the e-mail true completion all four recipients were already complete, so the three generated `write` continuations there would potentially damage a correct state (they were not executed). A random update on the same writers, with the same right factors and its per-writer BF16 perturbation norm matched to the learned one within 0.1%, leaves the screening contexts in place (FS+ 16/16, true completions 16/16, healthy 16/16): the effect is specific to the learned direction, a real causal lever on stopping, without selectivity on these contexts. At half the dose the edit gives 12/16 and 9/16: at both doses, true completions fall on these contexts as fast as or faster than false ones. A 25% dose was not run.

Two independent methods, a merge and a closed-form edit, thus move the decision. On the screening contexts, true completions fall at least as much as false ones in both, on two true-completion contexts; on the feasible cohort below, this holds for the merge, while the selectivity of the closed-form edit is not established. A simple argument makes a lever without a discriminator unsurprising. An edit applied at or after layer $\ell$ acts on a function of the residual stream at layer $\ell$ (all positions, and the recurrent state of the linear-attention layers), so it can treat a false and a true completion differently only to the extent that some function of that stream separates them, and to be useful that function must generalise to tasks it was not built on. Our probes, which read the stream at two positions, found no such separation beyond episode length (Table 1). A direct test on the writers that both methods edit finds the same. On 19 false and 19 true completions outside the reserve and the evaluation contexts, a ridge probe on the inputs (keys) of the 24 writers of layers 12–23, cross-validated by task, does no better than position. Averaged within test folds and weighted by their 71 positive–negative pairs, its median AUROC is 0.704 (range 0.61–0.90) at the end of the prefill and

0.634 at the end of reasoning, against 0.852 for the same probe on the step index; pooled out of fold, it is 0.596 and 0.496 against 0.644. This pool is weaker than it should be: 5 of its 19 false completions are infeasible tasks, on which `answer` is the correct tool, so the probe mixes in cases where the label marks no wrong decision, which weakens the argument. The strictly feasible cohort below avoids this for behaviour, though not for the probe. The two key subspaces are not identical (median principal angle 67–69°), but 83–86% of the false-completion key energy lies in the true-completion subspace. This is an empirical, non-total overlap together with a strong confound of position and length, not an impossibility: a low AUROC for a linear probe is not an information bound, the probes do not exhaust the stream, and even identical subspaces would not force a selective edit or merge to act globally, which would require proportional covariances.

The agent's own text carries no more. On 66 feasible false and 202 true successes from 90 tasks, the best of 7 detectors reading the final note, thought and reasoning, chosen after the fact, reaches a task-grouped AUROC of 0.66 (95% CI 0.55–0.76) against 0.72 (0.61–0.82) for the final step index; within the 18 tasks that end both ways, every detector lies between 0.46 and 0.54, and a within-task permutation test gives $p = 0.44$. Only the hesitation between samples carries a little: over four samples per context of the feasible cohort below, the original answers at a mean rate of 0.72 on false and 0.94 on true completions, an AUROC of 0.66 (0.53–0.80). We find the information neither in the activations nor in the agent's text, only in its hesitation from one sample to the next; at these sample sizes this is an absence of evidence, not a proof of absence.

**A strictly feasible cohort.** To take infeasible contexts out of the evidence, the original, the merge, the closed-form edit and its random control were re-measured on a cohort fixed before any output: 19 feasible false completions, 20 true completions and 16 healthy contexts, all from distinct tasks whose evaluators were read directly, with $k = 4$ paired seeds per context and an independent audit of every count. The original answers on 55/76 false and 75/80 true completions. The merge answers on 47/76 and 66/80: false answers fall by $-10.5$ points (95% CI $-25.0$ to 0.0), true ones by $-11.3$ ($-22.5$ to $-2.5$), a selectivity (first difference minus second) of +0.7 ($-16.1$ to +16.3). On feasible tasks, the merge is a lever without a discriminator. The closed-form edit answers on 42/76 and 68/80: false answers fall by $-17.1$ points ($-28.9$ to $-6.6$), on 9 of the 19 tasks rather than one context, and true ones by $-8.8$ ($-18.8$ to 0.0), a selectivity of $-8.4$ ($-22.6$ to +5.9). Its point estimate favours false completions, but the interval includes zero: neither a selective repair nor its absence is established. The drop survives removing the three tasks that overlap its construction or preservation sets ($-16.7$ points on 18 false completions, $-29.2$ to $-5.6$). Its random control, which left the screening contexts in place, here raises false answers by +15.8 points (+5.3 to +27.6) and leaves true ones unchanged (0.0), so the learned edit lies $-32.9$ points below its control on false completions ($-47.4$ to $-19.7$): its effect is specific to the learned direction. The random control is one case of a recurring pattern on this cohort: perturbations that degrade the model without targeting the decision raise false answers. Setting to zero the four attention heads ranked most selective in the attribution of §F.2 raises them by +21.1 points (+9.2 to +35.5; healthy valid actions $-3.1$, a screen not yet independently reviewed), and adding the full task vector of TMax-4B, an RL-trained Qwen3.5-4B, by +7.9 (+2.6 to +14.5; healthy valid actions $-4.7$, failing the healthy gate). Only the learned direction of the closed-form edit lowers them. A false "done" behaves like the default slope of a degraded model. All intervals are task bootstraps with 100,000 resamples, uncorrected for the four contrasts.

### F.6 Online evaluation

No candidate passed the gate, so none was compiled into the weights (Appendix O states the compilation step, Proposition 1); five arms were run online, each paired by task with an original run on the same harness, and the arms served by Modal with an original served by the same Modal endpoint (Table 3). The original itself is the first measure of noise: its three runs on the confirmation reserve, spread over 4.4 points (first row of Table 3), are the run-to-run noise against which every arm is read. It scores 43.2% on one harness and, on the other, 46.7% when served by the pod and 42.2% when served by Modal: the same weights differ by $-2.3$ points across harnesses (3 against 4 tasks better and worse) and by +4.4 points across servers (95% CI $-6.7$ to +17.8), differences comparable to the effects below, every one of which lies within a 95% CI that includes zero. Against it, no arm shows a gain: ExPO at $\lambda = 1.25$ $-2.3$ points (95% CI $-13.6$ to +9.1), the `answer` output-row edit 3/14 against 5/14, the corrective prompt 5/20 against 6/20, and, against the Modal-served original,

the CUA-Lite merge +6.6 (−6.7 to +22.1; 7 against 4 tasks better and worse) and the closed-form edit +2.2 (−13.3 to +17.8). Two reserve tasks score 0 under the pod-served original and 1 under both Modal-served arms; they score 0 under the Modal-served original as well, so the gap came from run-to-run variation, not from the server. The harnesses do differ: two `os` tasks score 0 on the local harness and 1 on the GitHub one, so runs from different harnesses are never pooled. The merge also produced 6 unparsable steps against none for the pod-served original, a sign that it degrades the output format. The decision rule for the merge was fixed before the comparison with the Modal-served original was read, but after the merge's own reserve scores had been seen, so only its TEST part is strictly prospective: the merge and an original served by the same Modal endpoint are compared on the 45 reserve and 120 TEST tasks, and a gain is claimed only if the lower bound of the task-bootstrap 95% CI of the paired difference is above zero. By the pre-registered rule (lower 95% bound above zero), the merge shows no detectable gain: over the 165 paired tasks it scores 92.5/165 against 89.3/165 ($\Delta$ +1.9 points, 95% CI −4.3 to +8.0; 18 against 14 tasks better and worse), and on the TEST split alone 70.5/120 against 70.3/120 ($\Delta$ +0.2, CI −6.5 to +7.5). On the same TEST tasks the pod-served original scores 72.1/120, +1.5 points above the Modal-served one (CI −4.5 to +7.6): the noise between two servings of the same weights is as large as the merge's effect. ExPO was also run on the TEST split without having passed the gate, a coordinator's decision we report as such: 70.3/120 against 72.1/120 ($\Delta$ −1.5 points, 95% CI −8.9 to +5.7; 11 against 12 tasks better and worse). All online scores are the official OSWorld score, a mean over tasks with partial credit included. With few tasks scoring differently between two runs (for the merge on the reserve, 7 against 4 better and worse), the bootstrap bounds move by up to a few points with the resampling seed; we report seed 0 throughout, and an independent recomputation agrees on every point estimate. Counting only full successes, the TEST difference for ExPO is −1.7 points (95% CI −9.2 to +5.8). On the reserve, a paired difference smaller than about 17 points would go undetected with 80% power, so these comparisons exclude only large effects. A cheaper paired design, branching from recorded episodes, was piloted: replaying a recorded episode up to its last step reproduces the screenshots in 9 of 10 validation episodes, and from the final step of 5 recorded episodes (two continuations each, both arms served by Modal) the original succeeds in 5/10 continuations and the merge in 4/10, with one episode whose replay diverges from its first step. The only failed episode that an original continuation recovers, `gimp/fbb548ca`, is an officially infeasible task, where the continuation states that the requested theme does not exist; on the feasible false successes whose replay is faithful, no continuation succeeds (original 0/2, merge 0/2). The pilot shows no advantage for the merge, no repair of a feasible false success, and is too small to show anything else.

# G Feci Guard: an external verifier

If the model does not separate a false "done" from a true one, the check has to come from outside, and it should judge the final state rather than the agent's account of it, as $\tau$-bench does by scoring the final database state [39]. Feci Guard is the simplest form of that check. When the agent calls `answer`, unless the answer declares the task infeasible, the guard takes a fresh screenshot and asks a separate verifier model to list every requirement of the task, to mark each one as met, unmet or not visible with its evidence, and to return a verdict. An INCOMPLETE verdict, which requires at least one requirement visibly unmet, refuses the answer and returns to the agent, as the answer's tool output, the missing requirement and a next action; the episode continues. A COMPLETE verdict, a timeout or an invalid output lets the answer through. At most two answers are refused per episode, and the third passes unchecked. A refusal always says what is missing, because blocking a false "done" without an explanation repaired none of the feasible false successes of the branching pilot (§F.6). The verifier (Claude Sonnet through the Claude Code CLI alias `sonnet`, whose exact served version was not recorded, a fixed prompt and JSON schema, a 55 s timeout) and the decision rule were frozen and registered before any evaluation.

**Offline, the verifier separates false from true successes.** On the same 268 episodes of the original agent, judged from the final screenshot only, the frozen verifier detects 68% of the false successes and rejects 10% of the true ones, a gap of +58 points (95% CI +41 to +74, bootstrap over 90 tasks; Table 4). Holo-3.1-4B judging the same episodes detects 53% and rejects 48% (gap +5, −13 to +23), not distinguishable from chance, and the paired difference between the two judges is +53 points (+26 to +79). A larger model of the same family, Holo4-27B with the same prompt as Holo-3.1-4B, rejects fewer true successes but detects no more false ones (gap +16, −3 to +33). With the verifier's

| Judge | false successes detected | true successes rejected | gap, points [95% CI] | no verdict |
|---|---|---|---|---|
| Holo-3.1-4B judging its own episodes | 35/66 (53%) | 97/202 (48%) | +5 [−13, +23] | 6 |
| Holo4-27B, same prompt | 31/66 (47%) | 63/202 (31%) | +16 [−3, +33] | 5 |
| Holo4-27B, frozen Feci Guard verifier | 31/66 (47%) | 49/202 (24%) | +23 [+5, +40] | 23 |
| Claude Sonnet, frozen Feci Guard verifier | 45/66 (68%) | 21/202 (10%) | +58 [+41, +74] | 0 |

**Table 4: An external verifier separates false from true successes; the agent judging itself does not.** The same 268 completed episodes of the original agent (66 feasible false successes, 202 true successes, 90 tasks), each judged once from the final screenshot only (the Holo models at temperature 0, the Claude verifier with the CLI's default settings). Detection is an INCOMPLETE verdict on a false success; rejection, one on a true success; a call with no verdict lets the answer pass and is counted in the last column. Gap: detection minus rejection, with a percentile bootstrap over tasks. The last two rows share the frozen Feci Guard prompt and rule (a JSON verdict per requirement, INCOMPLETE only if one is visibly unmet); the first two use a free-form prompt. An independent recomputation reproduces every count.

own prompt and rule, Holo4-27B detects 47% and rejects 24% (gap +23, +5 to +40; 23 calls without a verdict, counted as passes): +18 points above Holo-3.1-4B (−5 to +40) and −35 below Claude (−59 to −10). At equal prompt, Claude's advantage comes mostly from the model; the prompt helps Holo4-27B by lowering its false rejections from 31% to 24%, partly because its 15 calls without a verdict on true successes count as passes (26% on valid verdicts only). What a refusal is worth depends on the base rate. Among the tasks of the TEST split that the original, served by Modal, closes with `answer`, 27 score 0 and 65 score 1 (5 partial scores left out), a false share $f \approx 0.29$. With the verifier's detection $d$ and false rejection $r$ above, the precision of a refusal, $df/(df + r(1 - f))$, is about 0.73: roughly one refusal in four would hit a true success. This is an estimate, since the rates come from neighbouring distributions, not the same one. This is the measured basis for an external verifier, with three limits. One screenshot cannot show a saved file or a hidden setting, so detection cannot reach 100%. Each judge ran once. And Claude may have seen public OSWorld tasks.

**A stronger judge.** Claude Opus, given the two screens before the answer and the list of actions, detects more false successes than the frozen verifier on a held-out half, 20/29 against 16/29 (+13.8 points, paired task-bootstrap 95% CI +3 to +30; few discordant pairs, so the interval is approximate), with false rejections that rise by +4.3 points (−5 to +15). The variant was selected on a development half and confirmed once on a held-out half drawn by task hash and committed before any test: on that half it detects 69% (20/29) and rejects 12.9% (15/116), against 55% (16/29) and 8.6% (10/116) for the frozen verifier, with a paired bootstrap over its 47 tasks. A strict prompt demanding file evidence was rejected on the development half: it detected 35/37 false successes but rejected 33/86 true ones. The stronger judge has not been evaluated online.

**Online, the guard halves false "done" without a measurable score change.** The guarded agent was run once on the 120 tasks of the TEST split, under a rule and an analysis script committed before its results were read, and paired by task with the original served by the same Modal deployment on the same runners and harness. Its false "done" rate, a final `answer` on a task scored 0, falls from 22.5% to 10.8% (−11.7 points, 95% CI −19.2 to −4.2), meeting the registered primary criterion (an upper 95% bound below zero; the two servings of the original differ by less than two points, 24.2% against 22.5%). Its OSWorld score goes from 58.6 to 60.2 (+1.6 points, −4.4 to +7.7), which does not meet the secondary criterion (a lower bound above zero): we claim no score gain. The guard repairs 8 tasks the original failed and loses 4 the original solved, within the original's own run-to-run noise (served twice, it flips 9 tasks from 1 to 0 and 5 from 0 to 1), while avoiding 19 false "done"; the registered usefulness rule (repairs at least as many as losses, and fewer false "done") is met (net 8 minus 4). The registration also planned a replication on the confirmation reserve and an online run of the escalation to Holo4-27B; neither was run. Two amendments were committed before any comparison: after the pod serving the agent failed, and before any guarded episode, the agent and the primary comparator moved to the Modal endpoint; and six tasks lost to an infrastructure failure were re-run once. Against the original served by the pod, the false "done" rate falls by −13.3 points (−20.8 to −5.8) and the score changes by +0.1 (−6.5 to +6.6).

**Replayed repairs.** In one end-to-end episode on a LibreOffice Impress task the original had failed with a false success, the guard refused the first "done" because the strike-through stopped mid-word, the agent followed the feedback and fixed the line, and the second "done" passed; the official checker still scored the task 0, because the remaining difference lies in the saved file, which the

screen does not show. The guard also runs in our side-by-side live demonstration of the original and the guarded agent. A repair measure replays 12 recorded feasible false successes of the original up to their final answer and continues each twice per arm, with and without the guard, for at most 15 new steps, on the same server at the same time, scored by the official evaluator. Its protocol section was written before the arms were launched but first committed when about six branches had already finished, so the registration is self-declared; the exclusion rule depends only on the replay prefix, which runs before the continuation. 9 episodes replay faithfully in all four branches (3 excluded), 18 branches per arm. With the guard 5 of 18 branches are repaired, without it 2 of 18. The paired difference per episode is +16.7 points (95% CI 0 to +44, bootstrap over episodes; with nine discrete differences, the upper bound varies between +39 and +44 with the bootstrap seed); 2 episodes are better with the guard and 0 worse, a sign test gives $p = 0.5$, and over all 12 episodes the difference is +12.5 (0 to +33): the effect cannot be told apart from zero. The guard refused 12 branches (16 refusals in all), and 3 of them ended repaired, 25% (Wilson 95% CI 9–53%), from two episodes: on `calc/1de60575`, 2/2 against 0/2 without the guard, the verifier saw that the pivot table sat on a sheet with the wrong name and the agent renamed it; on `writer/0b17a146`, 1/2. `impress/8979838c` succeeds in all four of its branches as soon as the agent answers: its recorded false success does not reproduce under replay, so it accounts for both repairs without the guard and two with it, and only 3 repairs are specific to the guard. The costs are visible too: 7 refused branches reach the step limit without answering again, 2 pass at the refusal cap while still wrong, and on 2 episodes the guard accepts a wrong answer as complete in every branch. What is measured is a refusal plus coaching, since the feedback contains the verifier's suggested next action, not refusal alone. The refusal sometimes helps the agent repair, without statistical evidence at this stage.

**Escalating to a stronger agent.** The same twelve recordings were replayed with a different policy: when the guard refuses the original answer, the episode is handed, with the answer and the verifier's feedback in its history, to a stronger agent, Holo4-27B, for at most 20 new steps; the control hands every episode to Holo4-27B at the same point, without the guard. This contrasts the whole gated handoff with the stronger agent alone, not feedback alone, and the protocol was registered before any Holo4 sample. On the 9 episodes whose replay is valid in all four branches, the official score is 1 in 11/18 branches with the guard against 2/18 without it (9 against 0 after a new action), a paired difference of +50 points (95% CI +22 to +78, bootstrap over episodes; 5 episodes better, 0 worse). The result holds with a stricter replay check (8 episodes, +50, +12 to +88) and with all 12 episodes (+33, +4 to +62). After an actual refusal, 9/11 branches are repaired after a new action. This is a replay measurement: escalation to Holo4-27B was not run online, and the change it predicts on the TEST split, +6 to +12 points depending on the unmeasured cost of a wrongly refused true success, is a prediction.

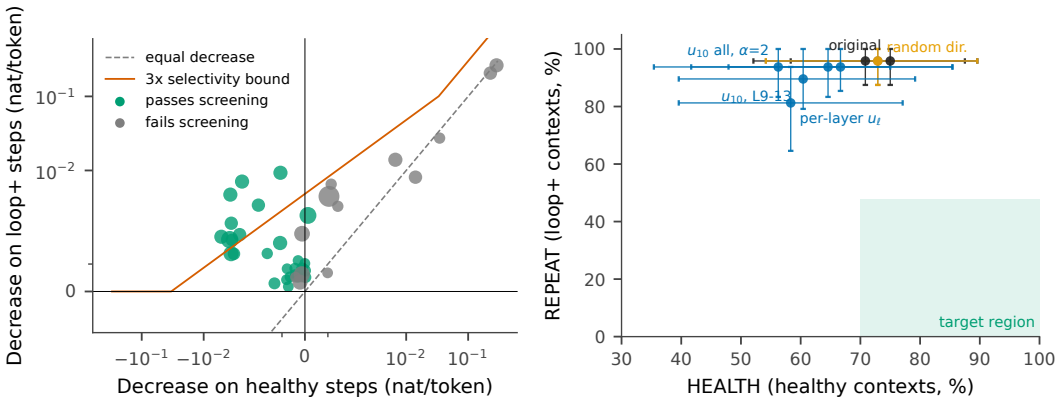
# H A first negative result: loops in Holo-3.1-0.8B

We first tried the recipe on the defect behind the weight-space programming result, looping, in Holo-3.1-0.8B on MiniWoB++. On 172 rollouts over 24 tasks the model succeeds on 40.7%; loops affect 24.5% of failures (106 LOOP+ steps), above the GO threshold.

**Detection works.** At layer 10, the loop direction reaches AUROC 0.893 (Fig. 2a). The position baselines reach 0.70 (step index) and 0.72 (context length), and an independent refit of the direction reaches 0.904.

**The weight edit is not selective.** Two rounds screened 40 weight edits along this direction (layers, writers, top-$k$ columns, $\alpha$ from 0.3 to 16; Appendix Table 14). The screening winner was frozen before the reserved seeds were read. It edits 1,953,863 weights (0.229% of the model). On reserved seeds it lowers the log-probability of loop actions by 2.31 mnat/token and that of healthy actions by 8.92 (ratio 0.26). A random direction under the same scope gives a ratio of 0.31, so it is *more* selective than the learned one (Table 11, Fig. 9a).

**Steering does not reduce loops.** In free generation, the original model repeats on 95.8% of loop samples and matches the healthy action on 75.0%. The pre-registered target was REPEAT $\leq 47.9\%$ with HEALTH $\geq 70.0\%$. The best arm, projecting each layer's own direction at all layers, reaches REPEAT 81.2% at the cost of HEALTH 58.3% (Fig. 9b, Table 12). The random direction




**(a)** Offline screening of 40 weight edits.

**(b)** Behavioural steering arms.

**Figure 9: On the 0.8B, edits large enough to matter hit healthy actions as hard as loops.** (a) Teacher-forced decrease in log-probability of the recorded action, loop versus healthy steps, on 40 fixed development contexts(symlog axes; marker area proportional to weights edited). Above the red line, candidates meet the screening selectivity bound, but only with negligible effects. (b) Loop rate versus healthy-action rate per steering arm, 95% bootstrap CIs over contexts. The shaded region is the pre-registered target, and no arm enters it.

(95.8%/72.9%) cannot be told apart from the identity hook (95.8%/72.9%). The loop direction is an unselective lever, not a usable one.

# I Discussion: why detecting is not causing here

**A marker of a state is not the cause of an action.** Looping is a property of the trajectory: the agent repeats because the screen did not change or the target was misread. A direction that encodes "this episode is stuck" can be read reliably while the repeated action is computed from features that do not depend on it. The global-erasure control shows this directly. Probes in computer-use agents can reach very high AUC without detecting the intended property [19], and format-confounded probes can lack any functional role [26]. Our 0.8B result is a case where the probe does track the intended property and still offers no lever.

**False success looks like a belief that the task is done, acted on without checking.** The 4B evidence points one way. The model says "done" on most false completions as on true ones, and its reasoning writes the conclusion first in both cases, a property of its output format (§C.1). The tool choice becomes visible late, at the tool name, after a thought that already declares success, through a circuit that reads the tool definition and treats true and false completions the same (§F.1). Pushing toward low value makes the model write that it should check, without making it check. This matches reports that stated confidence tracks commitment more than correctness [15] and that stopping decisions are spread over late components [7]. The perceived-completion direction (§F.4) adds that, on 14 matched pairs, its projection reads false completions as complete. The public report of an agent that claims success after acting with no screenshot at all [6] is the extreme case of the same pattern: the claim of completion is not conditioned on evidence that the task was done. Whether that state is a misperception of the screen or a commitment to finish, removing it is likely to suppress true completions too, and the FS− criterion of our gate exists to catch exactly that. The clearest single case argues against a pure perception error. On a Chrome rental task, the model reads the current date "Nov 8" on screen, then states that "Nov 10–11" is next month and answers "done"; a random direction of the same norm produces the same contradiction. The model looked; it compared wrongly. A second sample on the same task, under the "Wait" edit, goes further: it derives that next month is December, says the dates must be changed, then decides that the main requirement (sorting by seats) is met and answers "done". One output is not a population claim, but it suggests that a false "done" can be the partial satisfaction of a conjunctive task, with the unmet requirement noticed and then dropped. A repair may therefore have to enforce that every requirement holds, rather than act on looking or on the decision to stop.

**Implications for agent developers.** We found no direction whose removal lowers false "done" selectively, in screens able to detect only large effects; within our probes, nothing separated false from true completions beyond the episode's length except the hesitation between samples (Table 1, §F.5), and no edit that moved the decision showed an established selectivity on feasible tasks: the merge showed none, and the closed-form edit's interval includes zero. Our results suggest that a fix must bring in information the model does not use when it stops: an external verifier that checks each requirement of the task against the screen before `answer` is accepted (§G: offline, it separates false from true successes where the agent, judging itself with a free-form prompt, does not; online, it lowers the false "done" rate without a measurable score change), or training that targets false successes, for instance with rewards that penalise episodes ending in a false "done". Weight surgery of the kind tested here has not shown that it can supply it.

**What would change our mind.** The official analysis has already ruled out the first condition we set, a direction on at least 40 FS+ steps that beats position (Table 1). What remains is an intervention that moves FS+ answers while sparing FS− on unseen tasks, perhaps at the larger doses the ceiling effect calls for, and a paired online gain on the confirmation reserve larger than the 17 points our reserve can detect. Measuring how many false-success states can still be repaired when `answer` is forbidden would bound what any such edit can gain; we have not measured it.

# J Related work

**Directions, steering and ablation.** Behaviour-specific directions, built from contrasts of activations (differences of means, a single contrast pair, or principal components), are added at inference [18, 24, 33, 45] or projected out [2]; linear structure has also been reported for truthfulness [21]. Refusal in chat models is mediated by a single direction, and ablating it everywhere equals orthogonalising every writer [2], the equivalence of §O. Venhoff et al. steer backtracking and uncertainty in thinking models [34]. Weight-space programming removes a looping habit by editing `down_proj` columns [31].

**Agents, tools and termination.** Tool identity is linearly readable and steerable in single-turn settings, with steering inside agent loops left untested [36]. A direction can set the rate of tool calls [4], agents encode when to call a tool before reasoning [30], and steering has served as procedural memory for agents [42]. Huang et al. study when GUI agents should not act [12]. Models carry a value axis for being on the right track [13]; confidence tracks commitment [15] and supports self-detection of errors [16]; overconfidence has a mechanistic account [43]; and the decision to stop thinking runs through late MLPs [7]. On the evaluation side, $\tau$-bench judges an agent by the final state of the environment rather than by its account of the task, and measures reliability over repeated trials with pass$^k$ [39]; Feci Guard (§G) applies the first idea at run time, on the screen.

**Pitfalls.** A probe shows decodability, not use [3]; control tasks [11] and confound residualisation [26] are needed. Steering vectors generalise unevenly [28, 32], can compromise safety [14], push activations off the natural manifold [23], and are often beaten by prompting [37]. Patching has its own best practices [10, 41] and is the main tool of circuit discovery [35], and weight editing goes back to causal tracing [22].

# K Limitations

- **Small causal samples.** Each 4B screen uses six contexts from five tasks, and the reasoning analysis uses four tasks. These are screens for a lever, not estimates of effect size. A small effect on other tasks could have been missed, and we report "no admissible effect found", not "no effect exists".
- **The metric targets the wrong action on infeasible tasks.** 9 of 29 false successes are claims of completion on infeasible tasks, where `answer` with a refusal is the correct tool; one such context enters several screens and the ceiling test. Appendix M gives every count with and without it.
- **Many uncorrected comparisons.** Every arm of Tables 2, 6 and 7, the merges and the closed-form edits was screened on the same few contexts, and 8 arms were run online, without correction for multiple comparisons; no intervention on the model is claimed to work, and none would

survive a correction. The one positive result, the guard's lower false "done" rate (§G), is the primary criterion of its own registration, from a single TEST run.

- **The test split was used by 5 online runs and the guarded agent** (§F.6, §G): the original (served twice), ExPO, which had not passed the gate, a partial corrective-prompt run, and the CUA-Lite merge under a decision rule fixed in advance.
- **The confirmation reserve was used outside the protocol** (§F.6). The protocol kept it to confirm a committed candidate; no candidate was admitted, and five non-admitted arms were run on it: ExPO, the `answer` output-row edit, the corrective prompt, the CUA-Lite merge and the closed-form edit. It is no longer an untouched set for a future candidate.
- **Self-registered rules, LLM auditors and an LLM coordinator.** Our registrations are timestamped entries in our own repository, not a public registry; the primary online rule was fixed after the merge's reserve scores had been read, so only its TEST part is strictly prospective. The semantic audits and "independent" reviews are LLM agents, not validated against human raters, and some protocol decisions (for instance ExPO on the TEST split) were taken by an LLM coordinator.
- **One model, one benchmark.** Every result concerns Holo-3.1-4B (and the 0.8B) on OSWorld (and MiniWoB++) with one harness; the mechanistic analyses are exploratory, select and estimate on the same few contexts, and use artificial teacher-forced inputs. The Claude verifier may have seen public OSWorld tasks. The judge was called through the Claude CLI alias `sonnet` (Claude Sonnet, version not pinned); the Holo4-27B weights are `Hcompany/Holo4-27B` from Hugging Face at revision `4de03cfe`, as recorded in our deployment manifests.
- **Exploratory direction analysis.** The official analysis passed the 40-positive gate, but the second pass stopped at 80 tasks, so it is labelled exploratory; the steering screens predate it.
- **Automatic labels.** The 1% screen-change threshold, the 15 px click radius and the refusal regex are design choices, fixed before analysis but not validated against human labels.
- **Harness.** 50 steps, three screenshots and a third-party public harness (RealtimeGUI), not H Company's, whose infeasibility rule is broader than the official OSWorld adapter's (§B); absolute scores are not comparable with published numbers, paired comparisons are. One correct refusal is scored 0 because the harness's refusal regex misses it, and we kept the official score.
- **One family of directions.** Difference-of-means, matched pairs, PCA, attribution-selected neurons and heads. Sparse autoencoders and multi-direction subspaces were not tested.
- **Shared compute.** Research jobs shared one GPU with the rollout server and paused on a latency flag; interrupted runs are disclosed and resumed exactly, never counted as results.

# L Reproducibility and how this work was done

Models: `Hcompany/Holo-3.1-0.8B` at revision `72da4c53` and `Hcompany/Holo-3.1-4B` at `8c88265a`. Every number, table and figure in this paper is produced by `paper/figs/*.py` from result files copied verbatim into `paper/data/`; each macro in `paper/generated/num_*.tex` names its source file, and `build.sh` rebuilds the PDF from scratch. Every 4B arm is recorded with its raw outputs, job file, intervention hash and budget ledger, and each result was audited on CPU by an agent that did not run it. The experiments were planned, run and audited by AI agents (Claude Code sessions and GPT-6 Astra through the Codex CLI) under human supervision, following the protocol described here.

# M The infeasible context: every count with and without it

One FS+ context, Chrome `480bcfea` at step 4, comes from an infeasible task: "I do not like the design of the new 2023 chrome UI. I want to keep using the original one. Can you disable the new 2023 version chrome UI for me?" The 2023 interface cannot be switched off, so the correct final action is `answer` with a statement of infeasibility. The model instead claims that it disabled the new interface, which is a false success, but a non-`answer` action on this context is not a repair, and $P(\texttt{answer})$ is the wrong metric there. The context enters most screens of Tables 2 and 7, the merges and the first context of the ceiling test. Table 5 gives every affected count with and without it. Three readings change. The three non-`answer` calls of H6 are all on this context, so on the feasible FS+ context H6 answers 8/8 like the original. The only `answer` calls of H3 at $\lambda = 1$ are on this context too (0/8 without it). And without it, the CUA-Lite merge at $\beta = 0.5$ answers on 3/8 feasible false completions, a larger drop than in §F.5, still with true completions at 2/16. A second infeasible task,

Writer `bb8ccc78`, is the other FS+ context of the `perceive` screens, and there the model's answer states that the task cannot be done, in words the harness rule misses: both FS+ contexts of those screens are infeasible, so the `perceive` rows, including their "excl." column, contain no feasible false completion. None of the other conclusions changes: the arms that leave FS+ at its original rate do so on both contexts.

| Source | Arm | fs+ `answer` (all) | fs+ `answer` (excl. infeasible) |
|---|---|---|---|
| Steering screens (Table 2) | original reference | 8/8 | 4/4* |
| | residual addition L16 | 8/8 | 4/4* |
| | residual addition L24 | 8/8 | 4/4 |
| | "conclusion" direction, L31 | 8/8 | 4/4* |
| | Value Axis, value, L24 | 8/8 | 4/4 |
| | Value Axis, commitment, L24 | 8/8 | 4/4 |
| | Value Axis, value, L18 | 8/8 | 4/4 |
| | Value Axis, commitment, L18 | 8/8 | 4/4* |
| | circuit C1, $\alpha = 0.5$ | 8/8 | 4/4 |
| | circuit C1, $\alpha = 1.0$ | 8/8 | 4/4 |
| | circuit, random control, $\alpha = 0.5$ | 8/8 | 4/4 |
| | perceive, $\alpha = 0.5$ | 5/8 | 1/4 |
| | perceive, $\alpha = 1$ | 4/8 | 1/4 |
| | perceive, $\alpha = 2$ | 5/8 | 1/4 |
| | perceive, $\alpha = 4$ | 5/8 | 1/4 |
| | perceive, random control, $\alpha = 4$ | 7/8 | 3/4 |
| | perceive reference contexts (original) | 6/8 | 2/4 |
| | verify, calibration baseline | 4/4 | 0/0 |
| | verify, dose 1 | 8/8 | 4/4 |
| | verify, dose 2 | 8/8 | 4/4 |
| | verify, random control, dose 1 | 8/8 | 4/4 |
| | verify, random control, dose 2 | 8/8 | 4/4 |
| | weight edits, vLLM reference | 8/8 | 4/4 |
| | "Wait" output row | 6/8 | 3/4 |
| | linear-attention retention $\times 0.8$ | 8/8 | 4/4 |
| | linear-attention retention $\times 1.25$ | 8/8 | 4/4 |
| | H4 audit: false-completion claims | 5/8 | 2/4 |
| Mechanism-level edits (Table 7) | original (re-served) | 16/16 | 8/8 |
| | H2 | 16/16 | 8/8 |
| | H3, $\lambda = 0.5$ | 16/16 | 8/8 |
| | H5, $\lambda = 0.5$ | 16/16 | 8/8 |
| | H6 | 13/16 | 8/8 |
| | H2, $\lambda = 2$ | 16/16 | 8/8 |
| | H9 | 16/16 | 8/8 |
| | H10, $s = 0.5$ | 16/16 | 8/8 |
| | H10, $s = 0$ | 8/16 | 8/8 |
| | H3, $\lambda = 1$ | 3/16 | 0/8 |
| | audit: original, false-completion claims | 16/16 | 8/8 |
| | audit: H6, false-completion claims | 13/16 | 8/8 |
| | audit: H2, false-completion claims | 16/16 | 8/8* |
| | audit: H3, $\lambda = 0.5$, false-completion claims | 16/16 | 8/8* |
| | audit: H5, $\lambda = 0.5$, false-completion claims | 16/16 | 8/8* |
| Donor merge | original (re-served) | 16/16 | 8/8 |
| | CUA donor, $\beta = 0.25$ | 15/16 | 7/8 |
| | CUA donor, $\beta = 0.5$ | 11/16 | 3/8 |
| | Fara donor, $\beta = 0.5$ | 16/16 | 8/8 |

Not included: `neurons` (task 480bcfea is a construction task, not evaluated); `ratio`, `mit2`, `short`, `tact` (no task id 480bcfea in their data files). * every fs+ output answered, so the split follows from the per-context sample count.

**Table 5:** FS+ `answer` counts with and without the infeasible context `480bcfea@4`, for every arm that includes it. *Split implied exactly by the arm's cohort, because every FS+ output answered. Sources whose files carry no task identifiers are listed at the bottom.

# N Direction-free and mechanism-level weight edits

This appendix details the weight edits summarised in §F.5: extrapolation of the fine-tune, and edits that target a mechanism rather than a direction. None moves the decision on a feasible false completion without moving true completions or breaking generation.

| Edit (no gradient, no data) | fs+ `answer` | fs− `answer` | invalid/truncated |
|---|---|---|---|
| Original (re-served) | 16/16 | 14/16 | 0/32 |
| ExPO $\lambda = 1.1$, language layers | 16/16 | 15/16 | 0/32 |
| ExPO $\lambda = 1.2$, language layers | 16/16 | 13/16 | 0/32 |
| ExPO $\lambda = 1.25$, language layers | 16/16 | 13/16 | 0/32 |
| ExPO $\lambda = 1.5$, language layers | 16/16 | 13/16 | 1/32 |
| ExPO $\lambda = 1.5$, attention only | 16/16 | 15/16 | 0/32 |
| ExPO $\lambda = 1.5$, MLP only | 15/16 | 10/16 | 0/32 |
| Holo $+ 0.5$ (Qwen3.5-4B $-$ Base) | 16/16 | 16/16 | 0/32 |

**Table 6: Direction-free weight edits leave false "done" unchanged.** Free generation on two false-completion and two true-completion steps, 8 samples each, vLLM with the production harness call. Exploratory pilot.

### N.1 Edits without a direction: extrapolation and the Qwen post-training vector

Direction-free weight edits do not depend on locating anything. Holo-3.1-4B descends from the post-trained Qwen3.5-4B, not from its base: in the language layers, the fine-tune moves the weights by 4.7% of their norm, against 7.8% from the base and 6.4% between Qwen's base and post-trained models, and H Company's delta is almost orthogonal to Qwen's post-training (cosine $-0.02$). We therefore tested model extrapolation, $W(\lambda) = W_{\text{Qwen}} + \lambda\,(W_{\text{Holo}} - W_{\text{Qwen}})$ [44], on all language layers, attention only or MLPs only, and the addition of half of Qwen's post-training delta, in the spirit of reasoning vectors [40]. None changes false answers: they stay at 16/16 in every arm but one (15/16, MLPs only), where true completions drop more (Table 6). At $\lambda = 1.5$ one output in 32 loops to the token cap. The edit at $\lambda = 1.25$ keeps text and JSON intact, and it runs online as an arm of Table 3, because only OSWorld measures whether actions improve.

### N.2 Mechanism-level edits change the text, not the tool call

Three more gradient-free edits target a mechanism rather than a direction. H2 resets the 48 memory-gate matrices of the 24 Gated DeltaNet layers to Qwen's; H3 moves the 96 language MLP matrices halfway to Qwen's, leaving attention, memory, vision and the output head untouched; H5 moves the input embedding of `</think>` halfway toward the double-newline row, with the output head untied and kept. Each was served by vLLM on separate hardware against a re-served original, with the same 8 seeds on the same 6 development contexts. Each reproduces the original's tool counts exactly: 16/16 false and 15/16 true completions answer, every healthy action stays valid, no output is invalid (Table 7). Only 2/51, 0/51 and 4/51 outputs are token-identical to their paired original, and 21/51, 1/51 and 39/51 up to the end of reasoning (Fig. 10). This does not by itself show that the edits reach the computation: we did not measure how often two runs of the unmodified model with the same seeds differ on a batched server, and the H5 edit acts only once `</think>` has been generated, so its identical reasoning prefixes are expected. What the table does show is that the tool call is robust to these edits.

One edit does move the tool call, and shows why a moved tool is not a correction. Scaling the attention scores of the eight full-attention layers by $s = 0.5$ (H6, a softer attention, 8 `q_norm` vectors) lowers false-completion answers to 13/16, while true completions answer 16/16 and healthy actions stay valid. All 3 non-`answer` calls fall on the infeasible context, where `answer` is the correct tool; on the feasible FS+ context H6 answers 8/8, like the original (Appendix M). A blind semantic audit, with original and edited outputs mixed, finds 0/16 useful actions, as for the original, and 13/16 outputs that still claim success (original 16/16); the 3 non-`answer` calls are unhelpful theme clicks (GTK or Classic) justified by an incorrect prediction that they will restore the old interface, and none is useful. The tool moved; the task did not get closer to done. The second pre-registered dose, $s = 0.25$, was stopped at its pilot: the only output runs to the 3,072-token cap without closing its JSON, which closes the family. Four last edits close the series. Doubling the H2 gate edit ($\lambda = 2$, past Qwen), moving all 401 language tensors 20% of the way to Qwen at once (H9, a coordinated restoration rather than one block at a time), and halving the share of the query gain carried by the 64 rotary coordinates of each full-attention layer, the norm of the gain vector kept (H10, $s = 0.5$), leave the decision in place: false completions answer 16/16, 16/16 and 16/16, true completions 15/16, 16/16 and 15/16, and the blind audit finds no useful action. Removing that share entirely (H10, $s = 0$) repeats H6. False-completion answers fall to 8/16, all on the infeasible context, where

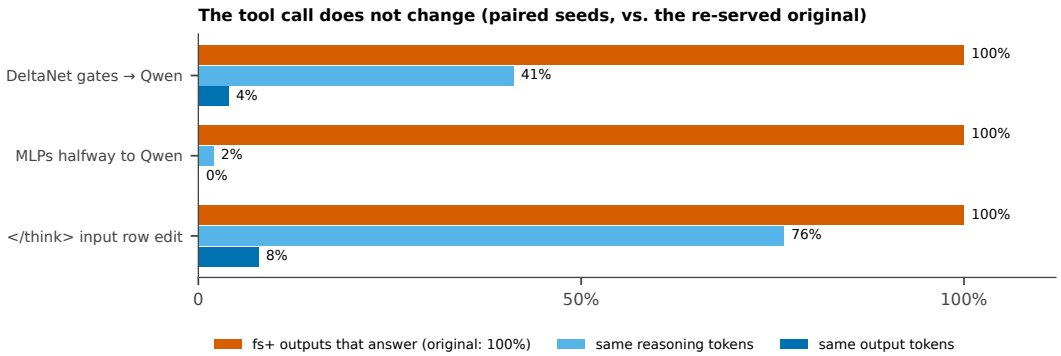



**Figure 10: The tool call does not change under three mechanism-level edits.** For three weight edits, share of false-completion outputs that still call `answer` (as the original does on all of them), and share of all 51 paired outputs whose reasoning, or whole output, is token-identical to the re-served original with the same seed. Six development contexts, 8 seeds each, plus three text prompts. Token identity has no null reference: two runs of the unmodified model were not compared.

7 theme clicks and 1 wait replace an answer that should have explained the impossibility; the feasible context still answers 8/8, and 2 healthy steps end in truncated outputs, a loss of at least 12.5 points that rejects the edit.

The same picture comes from three other directions. Seven two-neuron conditional circuits (15,360 weights each), written into the MLP of layer 19 or 31 to fire on an imminent conclusion and push toward verification, give 35/35 false answers; the layer-31 version inserts one "Wait" in every output, after which the model sometimes recognises the wrong month on Budget and rationalises it. Even the input such a circuit aims for does not help: injecting a generic instruction to compare each requested detail with the visible evidence into the reasoning, with the original weights, late or early, gives 5/5 and 5/5 false answers. A narrow MLP edit adapted from the weight-space programming recipe (64 columns in each of layers 17 and 20, 327,680 coefficients targeted) gives 8/8, 8/8, 8/8, 7/8 premature answers at $\alpha = 0.5$, 1, 1.5 and 6, and 0 useful actions in any arm; a random direction at $\alpha = 6$ gives 7/8, and true completions stay at 9–10/12 against 10/12. The last hypotheses closed the same way. Promoting the end-of-thinking token so that the model reasons less leaves `answer` rates unchanged at the dose that keeps outputs valid (false completions 13/20 against 13/20, true completions 18/20 against 18/20, on a 28-context panel), and breaks the JSON at higher doses (2/20 and 2/4 invalid outputs before the arms were stopped). In 3 further edits (penalising completion words, amplifying three task-reading heads, interpolating late MLPs toward Qwen), every dose gives 0/8 useful actions. An adaptation of TACT [29], which steers overthinking and overacting at the end of reasoning, could not be built as specified: after a blind audit of the labels, 13 overthinking and 7 overacting states remain, and the overthinking axis (AUROC 0.983) barely beats position and reasoning length (0.949). Pushing the MLP edit all the way breaks the model rather than the decision: with the MLPs set exactly to Qwen's, 34/48 outputs are invalid (its only `answer` calls are on the infeasible context), and one pilot output repeats a single token over 91.6% of its 3,072 tokens.

These screens share the limits of §F.3: 16 samples per stratum come from two contexts, the circuit arms stopped after 5 outputs once the gate could no longer be met and never measured protected classes, and the MLP edit is an adaptation of the published recipe, not a numerical reproduction.

| | tool calls | | | | blind audit, fs+ | | identical to original | |
|---|---|---|---|---|---|---|---|---|
| Weight edit | fs+ `answer` | fs− `answer` | healthy valid | invalid | false claim | useful | tokens | reasoning |
| Original (re-served) | 16/16 | 15/16 | 16/16 | 0/48 | 16/16 | 0/16 | – | – |
| DeltaNet gates → Qwen (H2, 48 matrices) | 16/16 | 15/16 | 16/16 | 0/48 | 16/16 | 0/16 | 2/51 | 21/51 |
| MLPs halfway to Qwen (H3, 96 matrices) | 16/16 | 15/16 | 16/16 | 0/48 | 16/16 | 0/16 | 0/51 | 1/51 |
| `</think>` input row → newline (H5, $\lambda = 0.5$) | 16/16 | 15/16 | 16/16 | 0/48 | 16/16 | 0/16 | 4/51 | 39/51 |
| Full-attention temperature $\times 0.5$ (H6, 8 vectors)$^\dagger$ | 13/16 | 16/16 | 16/16 | 0/48 | 13/16 | 0/16 | – | – |
| DeltaNet gates, $\lambda = 2$ (H2, 48 matrices) | 16/16 | 15/16 | 16/16 | 0/48 | 16/16 | 0/16 | – | – |
| Language layers 20% toward Qwen (H9, 401 tensors) | 16/16 | 16/16 | 16/16 | 0/48 | 16/16 | 0/16 | – | – |
| RoPE share of the query gain $\times 0.5$ (H10, 8 vectors) | 16/16 | 15/16 | 16/16 | 0/48 | 16/16 | 0/16 | – | – |
| RoPE share of the query gain $= 0$ (H10)$^\dagger$ | 8/16 | 13/16 | 14/16 | 3/48 | 8/16 | 0/16 | – | – |
| MLPs = Qwen (H3, $\lambda = 1$) | 3/16 | 3/16 | 8/16 | 34/48 | – | – | – | – |

**Table 7: Mechanism-level weight edits change the tokens, not the outcome.** vLLM 0.23 on separate hardware, same 6 contexts and 8 seeds per arm; the original is re-served on the same backend. The blind audit reads each false-completion output and counts claims of success and useful GUI actions, whatever tool is called. "Identical to original" counts the 48 GUI and 3 text outputs equal to the paired original, in full or up to the end of reasoning. One of the two FS+ contexts is an infeasible task, where `answer` explaining the impossibility is the correct action (Appendix Table 5). $^\dagger$Moving the tool is not a correction: every non-`answer` call falls on the infeasible context and is an unhelpful action justified by an incorrect prediction that it will restore the old interface (H6: 3 theme clicks; H10, $s = 0$: 7 theme clicks and 1 wait), none admitting the impossibility; none is useful. H10 at $s = 0$ is also rejected for truncated outputs on healthy steps, a loss of at least 12.5 points against a 10-point limit. Development pilot.

# O Compiling an intervention into the weights

No steering candidate passed the gate, so none was compiled; the weight edits of §F.5 were built directly by their own constructions. We state the compilation step because it constrains which steering interventions are worth testing. Let the residual stream at layer $\ell$ be $x_\ell = e + \sum_{j<\ell}(a_j + m_j)$, where $a_j = W_j^O z_j^a$ and $m_j = W_j^{\text{down}} z_j^m$ are the attention and MLP writes and $e$ the token or image embedding, and let $P = I - uu^\top$.

**Proposition 1** (Directional ablation equals writer orthogonalisation; cf. [2])**.** *In a pre-norm decoder whose sublayers read the residual stream only through their normalisation and whose writers have no bias, replacing every writer $W \in \{W_j^O, W_j^{down}, W^V\}$ by $PW$, the embedding by $W_E P$ and the vision-merger bias by $Pb^V$ gives a forward pass identical to the original model with $x \leftarrow Px$ applied after every write, at every layer and position.*

*Proof sketch.* By induction over writes: if $x$ has no $u$-component, every normalised sublayer input is unchanged, so every $z$ is unchanged, and each new write $PWz = P(Wz)$ keeps the sum orthogonal to $u$. □

Four consequences follow. A hook at block outputs is not equivalent, because the MLP then reads the un-ablated attention output. $W \leftarrow (I - \alpha uu^\top)W$ with $\alpha \neq 1$ scales each write's component by $1-\alpha$, whereas a residual hook compounds over layers. Tied embeddings (tied in both models) mean that projecting $W_E$ also projects the output head unless it is untied. And an addition $h \leftarrow h+\beta u$, like every intervention in Table 2 except projections, can be compiled only as a bias, which the bias-free projections of Holo-3.1 do not have. The circuit arm is the exception: scaling heads and neurons is exactly a column scaling of $W^O$ and $W^{\text{down}}$, 1,991,680 weights or 0.0439% of the model, and it has been tested in generation with no effect. Any compiled patch must reproduce the steered behaviour within 5 points per stratum before serving. The serving path is ready: a patched checkpoint is loaded by vLLM in about seven minutes, a null patch changes 0/724 tensors and reproduces the original outputs. Since no candidate passed, no compiled edit or equivalence check was run.

# P OSWorld diagnostic by domain

Table 8 breaks down the categories of §C by OSWorld domain.

| Domain | Tasks | Success | FS | Step lim. | Loop | Other + unsc. |
|---|---|---|---|---|---|---|
| multi-apps | 31 | 13/30 | 6 | 9 | 0 | 2 |
| chrome | 15 | 9/15 | 4 | 1 | 1 | 0 |
| lo-calc | 15 | 8/15 | 5 | 1 | 1 | 0 |
| lo-impress | 15 | 10/15 | 2 | 1 | 1 | 0 |
| gimp | 9 | 4/9 | 4 | 1 | 0 | 0 |
| lo-writer | 8 | 5/8 | 1 | 1 | 1 | 0 |
| os | 8 | 4/8 | 2 | 2 | 0 | 0 |
| vs-code | 8 | 6/8 | 1 | 0 | 1 | 0 |
| vlc | 6 | 2/6 | 4 | 0 | 0 | 0 |
| thunderbird | 5 | 5/5 | 0 | 0 | 0 | 0 |
| Total | 120 | 66/119 | 29 | 16 | 5 | 2 |

**Table 8:** Diagnostic split by domain (Holo-3.1-4B, pass 1). FS: false success; Other + unsc.: unclassified failures plus unscored tasks; lo: LibreOffice.

# Q Value-axis construction

The value direction of `astra-value` contrasts success with failure steps of the same relative stage, averaged within task, on 68 construction tasks; the commitment direction contrasts `answer` with continuation steps. Neither uses the FS+/FS− labels. Evaluated on FS+ versus FS− under task-grouped cross-validation, the value direction scores below 0.5 at layers 18, 24 and 28, so it is higher on true completions, while step index and trajectory length reach about 0.85. Its separation is thus weaker than position, the same confound as Trap 1.

# R Holo-3.1-0.8B on MiniWoB++: details

The tables below give the 0.8B study of §H in full: the diagnostic (Table 9), the offline screening of weight edits (Tables 11 and 14), the steering arms (Table 12) and the per-task results (Table 13).

| Primary category | Rollouts | % of failures |
|---|---|---|
| False success | 37 | 36.3 |
| Loop | 25 | 24.5 |
| Grounding miss | 21 | 20.6 |
| Abandon / budget | 10 | 9.8 |
| Wrong final action | 9 | 8.8 |
| Total failures | 102 | 100.0 |

**Table 9:** Primary failure category, 0.8B (102 failed rollouts).

| Layer | Contexts | min cosine | max rel. error |
|---|---|---|---|
| 0 | 40 | 0.99995 | 0.0104 |
| 10 | 40 | 0.99951 | 0.0314 |
| 23 | 40 | 0.99694 | 0.0787 |

**Table 10:** Replay fidelity on MiniWoB++: replayed vs. recorded residuals.

| Split | Steps | $n$ | Decrease | 95% CI (task bootstrap) | Random dir. |
|---|---|---|---|---|---|
| Dev (seeds 0–5) | loop+ | 72 | 3.89 | [1.32, 6.41] | – |
| | healthy | 344 | 9.97 | [6.62, 13.69] | – |
| | loop− | 287 | 9.16 | [6.51, 11.63] | – |
| | success | 119 | 10.86 | [3.99, 20.58] | – |
| Reserved (seeds 6–7) | loop+ | 23 | 2.31 | [-0.67, 7.62] | 1.30 |
| | healthy | 119 | 8.92 | [5.07, 14.03] | 4.19 |
| | loop− | 90 | 10.33 | [6.73, 14.81] | 4.20 |
| | success | 54 | 9.42 | [2.68, 19.39] | 5.33 |

**Table 11:** Frozen 0.8B weight edit vs. original: mean decrease in log-probability per token of the recorded action (mnat/token), task-level bootstrap CIs. The random-direction column uses the same layers, writers, $\alpha$ and top-$k$.

| Arm | REPEAT % [95% CI] | HEALTH % [95% CI] | Gens | Status |
|---|---|---|---|---|
| Original (no hook) | 95.8 [88, 100] | 75.0 [58, 90] | 96 | complete |
| Original, re-run | 95.8 [88, 100] | 70.8 [52, 88] | 96 | complete |
| Hook, $\alpha = 0$ | 95.8 [88, 100] | 72.9 [54, 90] | 96 | complete |
| Random dir., all layers | 95.8 [88, 100] | 72.9 [54, 90] | 96 | complete |
| $u_{10}$ at L10 | 93.8 [85, 100] | 66.7 [48, 85] | 96 | complete |
| $u_{10}$ at L9–13 | 89.6 [79, 100] | 60.4 [40, 79] | 96 | complete |
| $u_{10}$, all layers | 93.8 [83, 100] | 64.6 [42, 85] | 96 | complete |
| $u_{10}$, all layers, $\alpha = 2$ | 93.8 [83, 100] | 56.2 [35, 75] | 96 | complete |
| per-layer $u_\ell$, all layers | 81.2 [65, 94] | 58.3 [40, 77] | 96 | complete |
| matched-pair dir. | n/m | $\leq 66.7$ | 36 | rejected early |
| matched-pair PCA | n/m | $\leq 62.5$ | 36 | rejected early |
| MLP-site dir. | n/m | $\leq 66.7$ | 36 | rejected early |
| MLP-site PCA | n/m | $\leq 62.5$ | 40 | rejected early |
| mean of layer dirs. | n/m | $\leq 64.6$ | 40 | rejected early |
| strict pairs | n/m | $\leq 62.5$ | 36 | rejected early |
| strict pairs PCA | n/m | $\leq 64.6$ | 40 | rejected early |
| strict MLP | n/m | $\leq 68.8$ | 36 | rejected early |
| strict MLP PCA | n/m | $\leq 66.7$ | 36 | rejected early |
| $u_{10}$ amplification $\alpha = -1$ | n/m | n/m | 4 | aborted (invalid) |
| constant addition $-0.25\, u_{10}$ | n/m | n/m | 4 | aborted (invalid) |

**Table 12:** 0.8B free-generation steering arms (12 loop + 12 healthy contexts, $k = 4$). "Rejected early" arms show an upper bound on HEALTH that already fails the threshold.

| Task | Success | Task | Success |
|---|---|---|---|
| `click-option` | 7/7 | `click-menu` | 2/7 |
| `click-tab` | 8/8 | `use-autocomplete` | 2/7 |
| `email-inbox-delete` | 7/7 | `click-tab-2` | 1/7 |
| `click-checkboxes` | 7/8 | `form-sequence` | 1/7 |
| `click-widget` | 7/8 | `login-user` | 1/7 |
| `navigate-tree` | 6/7 | `login-user-popup` | 1/7 |
| `click-dialog-2` | 6/8 | `multi-layouts` | 1/7 |
| `focus-text-2` | 3/7 | `social-media` | 1/7 |
| `enter-text-2` | 3/8 | `click-collapsible` | 0/7 |
| `click-checkboxes-soft` | 2/6 | `enter-date` | 0/7 |
| `choose-list` | 2/7 | `enter-password` | 0/7 |
| `click-button-sequence` | 2/7 | `search-engine` | 0/7 |

**Table 13:** 0.8B success per MiniWoB++ task.

| Candidate | Layers | $\alpha$ | top-$k$ | Writers | Weights | $D_{\text{loop+}}$ | $D_{\text{healthy}}$ | $D_{\text{loop}-}$ | $D_{\text{succ}}$ | Pass |
|---|---|---|---|---|---|---|---|---|---|---|
| r3_distributed_a16 | 9–23 | 16.0 | 64 | mlp+attn | 1,966,080 | 262.66 | 286.96 | 259.44 | 342.09 | no |
| r3_early_a16 | 9–10 | 16.0 | 256 | mlp+attn | 1,048,576 | 206.20 | 228.26 | 197.14 | 284.70 | no |
| r3_L10_a16 | 10 | 16.0 | 64 | mlp+attn | 131,072 | 27.34 | 34.61 | 31.67 | 43.46 | no |
| r3_distributed_a4 | 9–23 | 4.0 | 64 | mlp+attn | 1,966,080 | 13.96 | 6.69 | 8.61 | 3.29 | no |
| a2_r3_a3 | 9–23 | 3.0 | 64 | mlp+attn | 1,966,080 | 9.31 | -1.07 | 2.23 | -6.32 | yes |
| r3_early_a4 | 9–10 | 4.0 | 256 | mlp+attn | 1,048,576 | 8.11 | 14.12 | 8.68 | 17.66 | no |
| a2_r3_a25 | 9–23 | 2.5 | 64 | mlp+attn | 1,966,080 | 7.08 | -2.74 | 0.18 | -7.62 | yes |
| r3_mit_layers_a16 | 15,19 | 16.0 | 64 | mlp | 131,072 | 6.53 | 1.14 | -1.04 | 2.85 | no |
| a2_r2_distributed_a2 | 9–23 | 2.0 | 64 | mlp+attn | 1,966,080 | 4.72 | -3.71 | -1.26 | -8.14 | yes |
| r2_distributed_None | 9–23 | 1.0 | all | mlp+attn | 86,507,520 | 4.47 | 1.05 | 1.89 | 0.25 | no |
| a2_r3_mlp | 9–23 | 2.0 | 64 | mlp | 983,040 | 3.40 | -2.03 | 0.55 | -5.25 | yes |
| r3_L10_a4 | 10 | 4.0 | 64 | mlp+attn | 131,072 | 3.28 | 1.43 | 0.66 | 1.70 | no |
| r2_early_None | 9–10 | 1.0 | all | mlp+attn | 11,534,336 | 2.77 | 0.13 | 0.32 | -1.45 | yes |
| a2_r3_k32 | 9–23 | 2.0 | 32 | mlp+attn | 983,040 | 2.49 | -3.59 | -2.42 | -6.75 | yes |
| r1_L10_mlp_attn_None | 10 | 1.0 | all | mlp+attn | 5,767,168 | 2.11 | -0.13 | 0.73 | -1.22 | no |
| a2_r3_attn | 9–23 | 2.0 | 64 | attn | 983,040 | 2.07 | -2.85 | -2.16 | -4.49 | yes |
| r2_distributed_64 | 9–23 | 1.0 | 64 | mlp+attn | 1,966,080 | 1.99 | -5.19 | -3.24 | -8.60 | yes |
| r2_late_attn_None | 10,13,15,17,18,19,20 | 1.0 | all | attn | 14,680,064 | 1.89 | -3.80 | -2.32 | -6.17 | yes |
| a2_r3_k16 | 9–23 | 2.0 | 16 | mlp+attn | 491,520 | 1.88 | -3.50 | -2.93 | -5.92 | yes |
| r1_L10_attn_None | 10 | 1.0 | all | attn | 2,097,152 | 1.77 | -1.08 | -0.38 | -2.30 | yes |
| r3_mit_layers_a4 | 15,19 | 4.0 | 64 | mlp | 131,072 | 1.38 | -1.64 | -1.08 | -1.82 | yes |
| a2_r2_distributed128 | 9–23 | 0.7 | 128 | mlp+attn | 3,932,160 | 1.38 | -3.61 | -3.23 | -5.62 | yes |
| r2_late_attn_64 | 10,13,15,17,18,19,20 | 1.0 | 64 | attn | 458,752 | 1.37 | -3.25 | -2.34 | -5.11 | yes |
| a2_r1_matched_mlp_attn | 10 | 0.5 | 64 | attn | 65,536 | 1.13 | -0.30 | -0.52 | -0.53 | yes |
| a2_r1_matched_resid_attn | 10 | 0.5 | 64 | attn | 65,536 | 1.02 | -0.00 | 0.33 | -0.45 | yes |
| a2_r2_matched_early | 9–10 | 0.7 | 128 | attn | 262,144 | 0.84 | -0.12 | -0.89 | -0.23 | yes |
| r1_L10_mlp_attn_64 | 10 | 1.0 | 64 | mlp+attn | 131,072 | 0.84 | -0.43 | 0.04 | -1.27 | yes |
| r1_L10_attn_64 | 10 | 1.0 | 64 | attn | 65,536 | 0.83 | -0.78 | -0.67 | -1.57 | yes |
| a2_r2_matched_late | 10,13,15,17,18,19,20 | 0.7 | 64 | attn | 458,752 | 0.76 | -0.01 | 0.13 | -0.45 | yes |
| a2_r1_mlp_attn | 10 | 0.5 | 64 | attn | 65,536 | 0.68 | 0.99 | 0.71 | 1.08 | no |
| a2_r2_matched_attn16 | 10 | 0.3 | 16 | attn | 16,384 | 0.63 | -0.44 | -0.29 | -1.30 | yes |
| r2_mit_layers_None | 15,19 | 1.0 | all | mlp | 7,340,032 | 0.63 | -0.14 | 0.53 | -0.11 | no |
| r1_L10_mlp_64 | 10 | 1.0 | 64 | mlp | 65,536 | 0.53 | -0.65 | -0.53 | -1.14 | yes |
| r2_mit_layers_64 | 15,19 | 1.0 | 64 | mlp | 131,072 | 0.53 | -0.30 | 0.05 | -0.54 | yes |
| a2_r1_matched_resid_mlp | 10 | 0.5 | 64 | mlp | 65,536 | 0.52 | -0.41 | 0.21 | -1.35 | no |
| a2_r2_matched_attn128 | 10 | 0.7 | 128 | attn | 131,072 | 0.52 | 0.03 | -0.49 | -0.19 | yes |
| a2_r1_mlp_mlp | 10 | 0.5 | 64 | mlp | 65,536 | 0.43 | -0.80 | -0.64 | -1.46 | yes |
| r1_L10_mlp_None | 10 | 1.0 | all | mlp | 3,670,016 | 0.34 | -0.21 | 0.14 | -0.71 | no |
| r2_early_64 | 9–10 | 1.0 | 64 | mlp+attn | 262,144 | 0.30 | -1.33 | -0.60 | -2.79 | yes |
| a2_r1_matched_mlp_mlp | 10 | 0.5 | 64 | mlp | 65,536 | 0.18 | -0.72 | -0.03 | -1.32 | yes |

**Table 14:** All offline weight-edit candidates on the 0.8B (40 fixed development contexts). $D$: decrease in mean log-probability per token of the recorded action (mnat/token). "Pass": $D_{\text{loop+}} > 0$ and $D_{\text{loop+}} \geq 3\max(0, D_{\text{healthy}}, D_{\text{loop}-}, D_{\text{succ}})$.

# NeurIPS Paper Checklist

1. **Claims**
   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   Answer: [Yes]
   Justification: The abstract and §1 claim a diagnosis, three measured traps, a causal record in which no intervention passes the gate selectively, a mechanistic account of that failure and an offline and online evaluation of an external verifier; §2–§6 and the appendices support each claim, and the abstract states that the screens are pilots that detect only a fall to 4/8 false answers on a single feasible false-success context, and that the verifier's single pre-registered TEST run lowers the false "done" rate with no measurable score change (no score gain is claimed).
   Guidelines:
   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.
2. **Limitations**
   Question: Does the paper discuss the limitations of the work performed by the authors?
   Answer: [Yes]
   Justification: §6 (Limitations) and Appendix K: small causal samples that exclude only large effects, infeasible screening contexts, a three-context ceiling test, an exploratory direction analysis, reuse of the reserve and TEST split by non-admitted arms, many uncorrected comparisons, self-registered rules (the primary online rule was fixed after the merge's reserve scores were read), automatic labels and LLM auditors not validated against humans, a third-party harness, possible donor and verifier contamination, a single online guard run with a proprietary verifier, and a single model and benchmark.
   Guidelines:
   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.
3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
Answer: [Yes]
Justification: The paper's one formal statement, Proposition 1 (directional ablation equals writer orthogonalisation), states its assumptions (a pre-norm decoder whose sublayers read the residual stream only through their normalisation, bias-free writers, every writer and the embedding projected); a proof by induction over writes follows it in Appendix O; it is short and given as a sketch, which we flag here.
Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?
Answer: [Yes]
Justification: Models and revisions, harness commit, sampling settings, frozen split hashes, the gate, seeds and every intervention's definition are given in §2–§4 and Appendices B–L; every number is regenerated by script from released result files.
Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Code, frozen splits, raw generations, audits and the scripts that rebuild every number, table and figure will be provided to reviewers as an anonymised supplementary archive; raw outputs are released upon publication. Models and benchmarks are public, except the Claude verifier and judges, a proprietary service whose prompts we give and whose exact served version was not recorded.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (https://neurips.cc/public/guides/CodeSubmissionPolicy) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (https://neurips.cc/public/guides/CodeSubmissionPolicy) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Sampling parameters, contexts, seeds, doses, layers and the selection procedure of every arm are in §3, §4, Table 2 and Appendix F. No model is trained.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Success rates and recurrence rates carry Wilson 95% intervals, each fs+ answer rate in Table 2 has its Wilson interval, and 0.8B results use task-level bootstrap intervals. We state that the intervals ignore clustering by task and that the screens can only detect large effects.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.

- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: We report the hardware (one NVIDIA H200 shared with the rollout server, rented H100 GPUs for serving edited checkpoints, a separate CPU host for the VMs; Appendix B) and the budget of forward passes and generation batches of each arm, with order-of-magnitude estimates of total compute (Appendix B): about 14 hours of one H200, shared by all research jobs and the rollout server, so per-experiment GPU-hours are not separable; about 8 H100 GPU-hours; about 60 CPU jobs for the OSWorld VMs.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The work uses public models and benchmarks, involves no human subjects or personal data, and reports negative results and limitations in full.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: §6 (Broader impact): false success claims mislead users and downstream automation; our results argue for external verification and caution with interpretability-based repairs. The recommended fix sends desktop screenshots to a third-party model, a privacy cost, and about one refusal in four would hit a true success (§G); we release no edited model.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.

- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: We release no model or scraped data; edited checkpoints were used for evaluation only.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: All assets are cited, with licences in Appendix B: Holo-3.1, Qwen3.5-4B, the CUA-Lite donor, OSWorld and BrowserGym are Apache-2.0 and MiniWoB++ is MIT; the RealtimeGUI harness has no licence file at the cited commit, which we state, and we use it unmodified except for the model provider and three VM-provider infrastructure settings (Appendix B). Holo4-27B, Fara1.5-4B and TMax-4B are used as published by their authors for evaluation only, and the Claude verifier and judges through Anthropic's service under its terms of use.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The released code and result files are documented: each generated macro names its source file, and the repository describes the pipeline, the protocol and the budget ledgers.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.

- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: No crowdsourcing or human subjects.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: No human subjects.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: Experiments were planned, run and audited by LLM agents (Claude Code sessions and Codex agents; Appendix L) under human supervision, some protocol decisions were taken by an LLM coordinator, and LLMs are method components (the Claude verifier and judges, and the semantic audits of changed actions), following the protocol of §3; every result was audited by an agent that did not produce it (Appendix L). LLMs also assisted in writing.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.